ARM: Access Request Management

VIRSA 4.0 ------ > Access Enforcer (AE)
GRC AC 5.1, 5.2, 5.3 ------- > Compliant User Provisioning (CUP)
GRC AC 10, 10.1,12-------> Access Request Management (ARM)

ARM will take care of the following:

- Automate Approvals and Documentation through **MSMP** Workflow
- Auto Provisioning – Automatic user maintenance

Advantages of Automation: AI (Artificial Intelligence) / RPA - Robotic Process Automation)

- No time delays.
- Reduction of human errors
- Enforce the Process to be followed - Workflow.

User Administration – User creation, user modification and User deletion
Process to be followed for user administration.

1. Users must fill in the SAP Access form.
2. It must be approved by his Reporting manager.
3. It must be approved by Role Owner (for the roles user is requesting)
4. Attach this form and raise a request in ITSM/Service Now – Ticketing tool.
5. Security team will create or modify or delete user as per the request. (Auto provisioning)

How can you ensure that the above process is not violated?

- Remove SU01 Access
- Automate user Creation

Automation & Workflow designing can be done for the following.

- Process of user administration – Non-HR & HR System (2) - **ARM**
- Role Approval (1) - **BRM**
- Function maintenance Approval (1) - **ARA**
- Risk Maintenance Approval (1) - **ARA**
- FF log report review (1) **- EAM**
- Review – SOD Review & User Access Review (2) - **ARA**
- Mitigation Control – Maintenance and Assignments Approvals -(2) **ARA**

Process IDs (Workflow types)

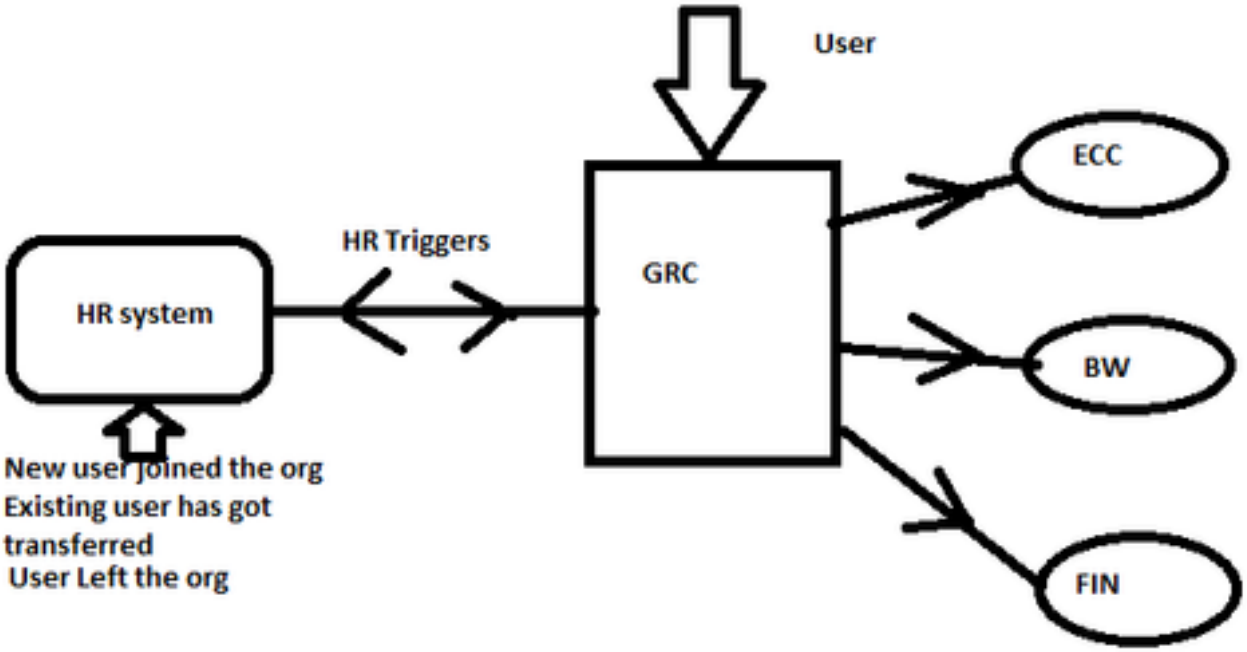
Select workflow process

Process ID	Process Description
SAP_GRAC_ACCESS_REQUEST	Access Request Approval Workflow
SAP_GRAC_ACCESS_REQUEST_HR	Access Request Approval for HR OM Objects Workflow
SAP_GRAC_CONTROL_ASGN	Control Assignment Approval Workflow
SAP_GRAC_CONTROL_MAINT	Mitigation Control Maintenance Workflow
SAP_GRAC_FIREFIGHT_LOG_REPORT	Fire Fighter Log Report Review Workflow
SAP_GRAC_FUNC_APPR	Function Approval Workflow
SAP_GRAC_RISK_APPR	Risk Approval Workflow
SAP_GRAC_ROLE_APPR	Role Approval Workflow
SAP_GRAC_SOD_RISK_REVIEW	SOD Risk Review Workflow
SAP_GRAC_USER_ACCESS_REVIEW	User Access Review Workflow

ARM configuration is done in the below path.

1. **Auto Provisioning**: GRC ----- > AC ----- > User Provisioning
2. **Workflow (MSMP)**: GRC ----- > AC ----- > Workflow for Access Control

<u>Maintain Settings for HR Trigger</u>



<u>User Details ---- > HR System</u>
<u>Designation --- > Roles</u>
<u>Request Reason --- > NEW USER</u>

HR Master is created --- Action ID 00001 -----> New Account
HR Master is modified --- Action ID 00002 -----> Change Account
HR Master is deleted/deactivated --- Action ID 00003 ------> Delete Account

Maintain Service Level Agreements (SLA):

- SLA can define based on Process ID (workflow Type)
- It can't be defined based on Request type

Define Request Type:

Correctly map request types to the respective Process IDs or workflow types
Ex,,
Access request approval workflow will have following request types.

- New user creation
- Change user
- Delete User
- Lock User

Risk approval workflow will have following request types

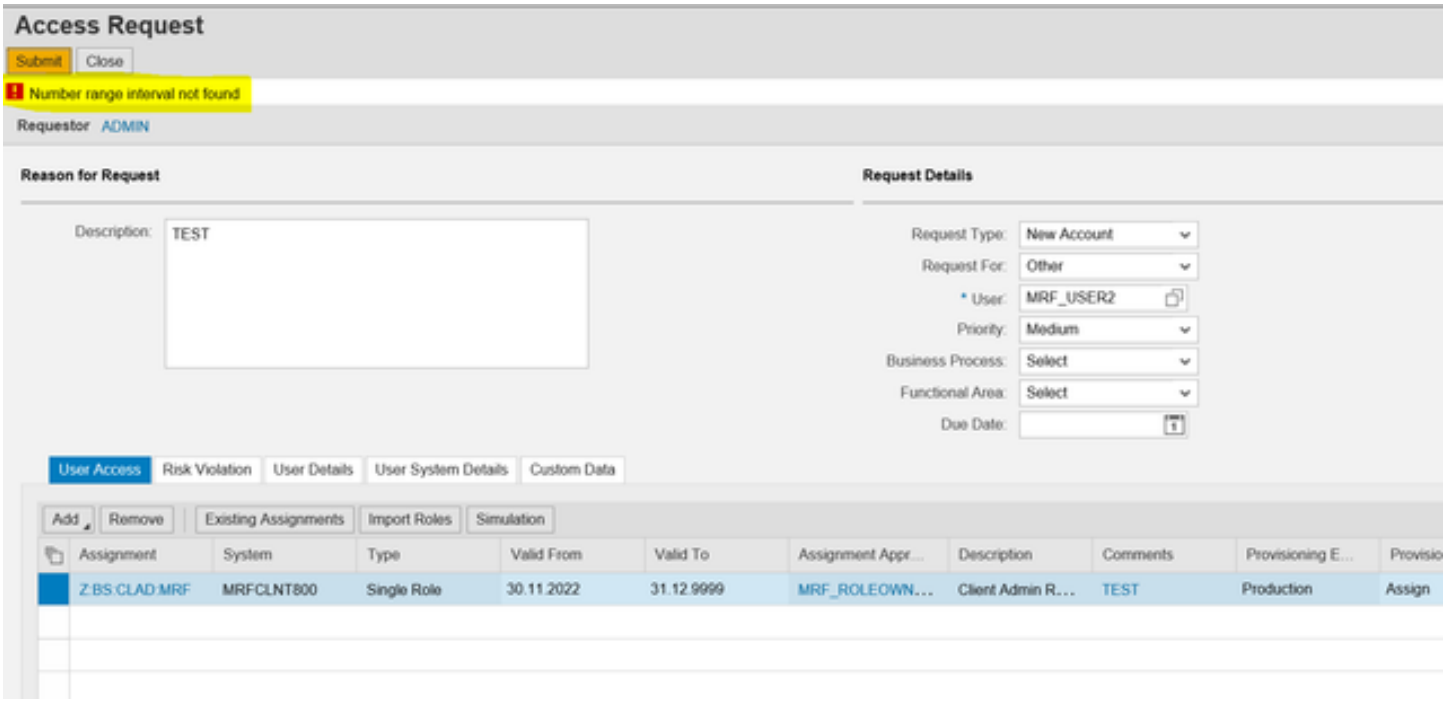
- Risk creation
- Risk Modification
- Risk Deletion

Function approval workflow will have following request types

- Function creation
- Function Modification
- Function Deletion

Maintain Number Range Intervals for Provisioning Requests:

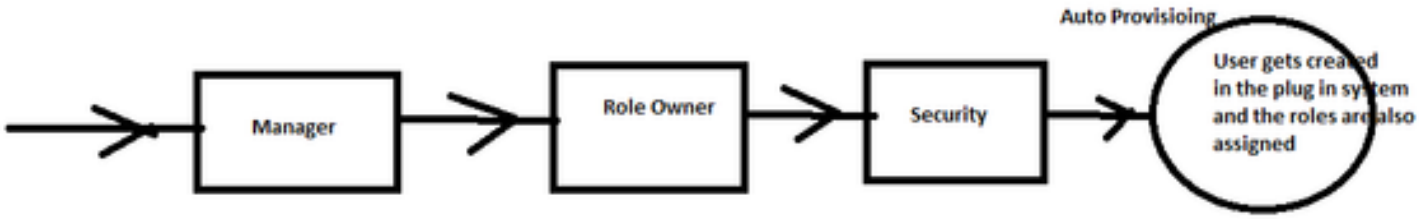
This is related to the Access request number (range) that is generated during submission of Access request by user.
Object: **GRACREQNO**
Number Ranges ----- > Modify Intervals ----- > Insert Interval



Define Number Range for Provisioning Requests:
Choose the appropriate number range as active
Maintain End User Personalization: (EUP)
Access Form customization
Access form Fields can be made as

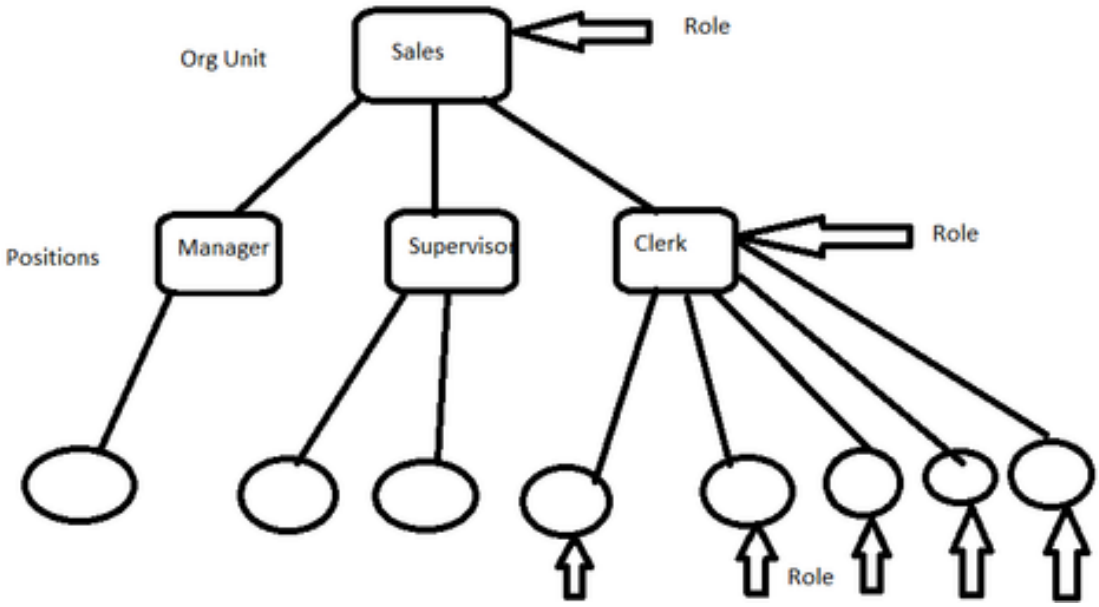
- Mandatory (YES/NO)
- Editable (YES/NO)
- Visible (YES/NO)

Maintain Provisioning Settings:



Role Provisioning Type

- Direct (SU01 role assignments)
- Indirect (HR system – Position or Org Unit based role assignments)
- Combined



Auto Provisioning
Select: Auto Provisioning at end of the request
What is difference between?

1. Auto Provisioning at the end of the request
2. Auto Provisioning at the end of the path

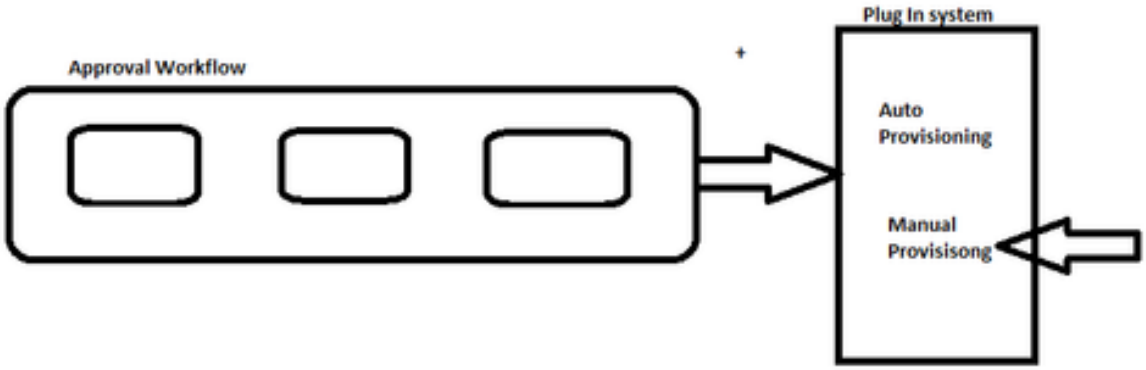
What is difference between?

1. Auto Provisioning – GRC system creates User
2. Manual Provisioning – Security Consultant created the User ID

What is the difference between No provisioning and Manual Provisioning?

No provisioning ---- > Request is closed after approvals are completed.

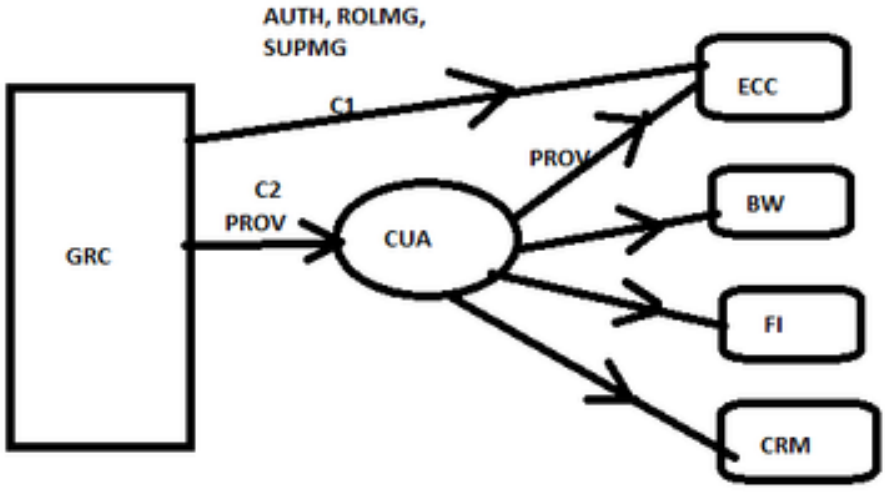
Manual provisioning ---- > Request is not closed after approvals. Manually we need to create user and assign roles, then manually we must close the request.



Maintain User Defaults:

This is related to DEFAULTS tab under SU01

CUA Settings



Activate End User Logon:

Role of End User in GRC System: (Not Approver)

1. Able to submit Access request

10K users in my Org

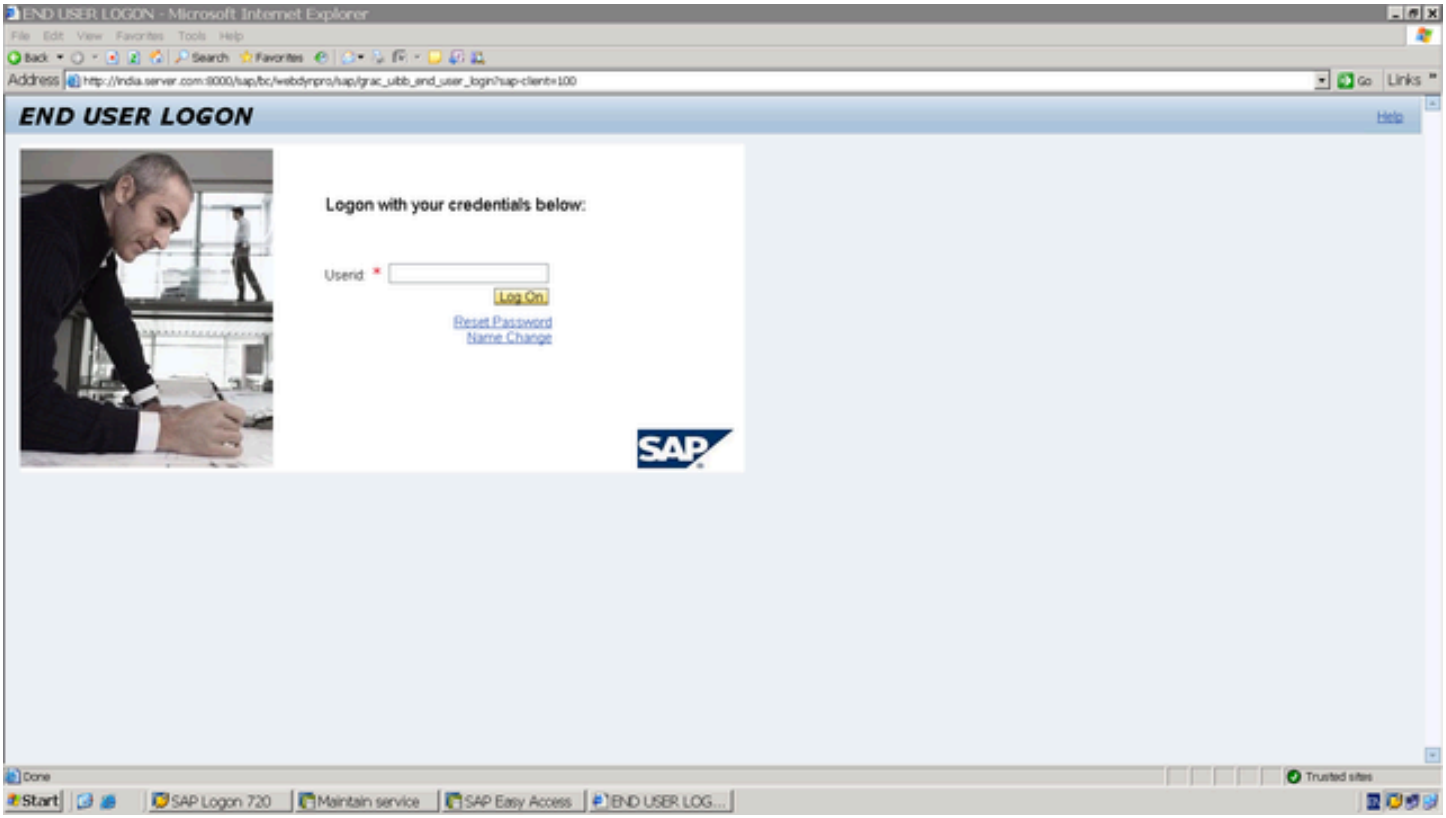
SAP used to charge for these 10K users - GRC License

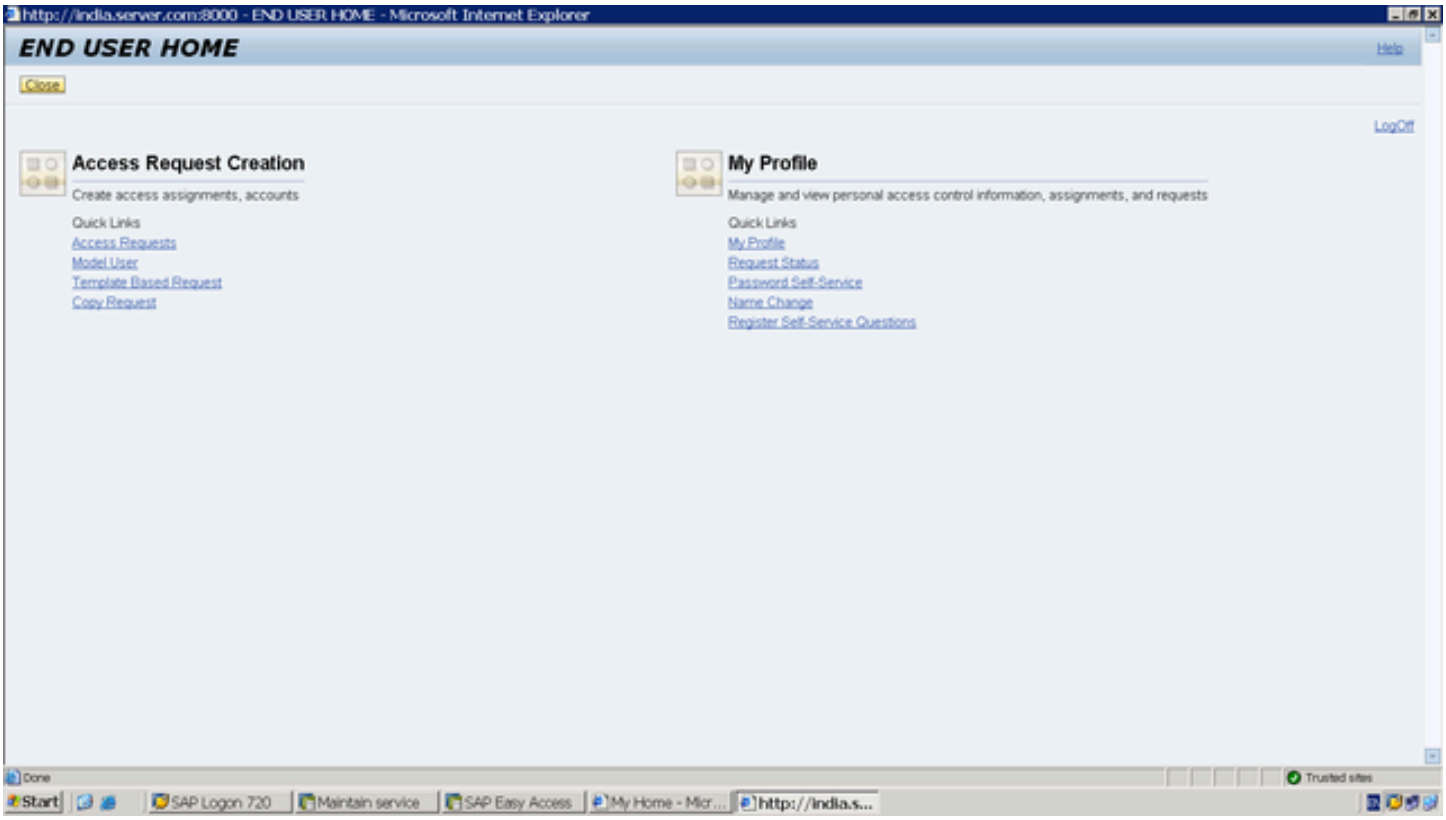
Solution:

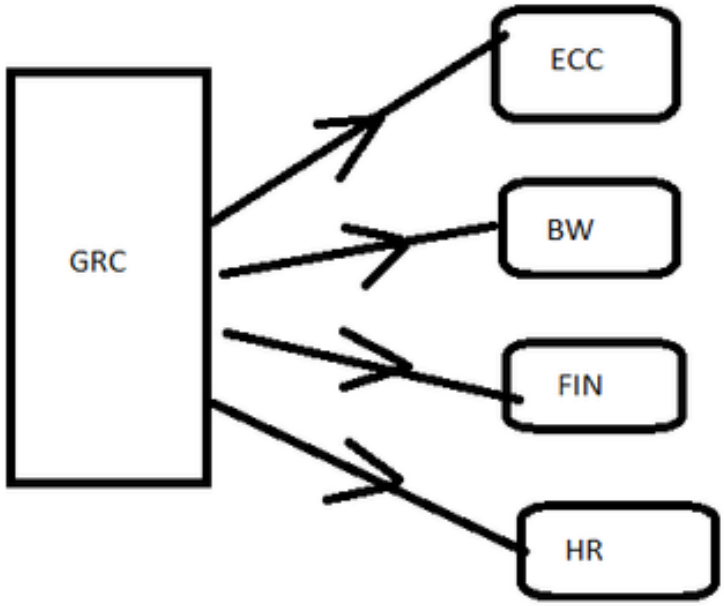
There is no need to create users in GRC and still users can submit access forms in GRC.
User should be present in some place (SAP/LDAP/AD/HR) – Verification/Authentication

End User Login:

http://india.server.com:8000/sap/bc/webdynpro/sap/grac_uibb_end_user_login?sap-client=100







- SLA can be defined at **Process ID level** only.
- Number Ranges Object is **GRACREQNO**

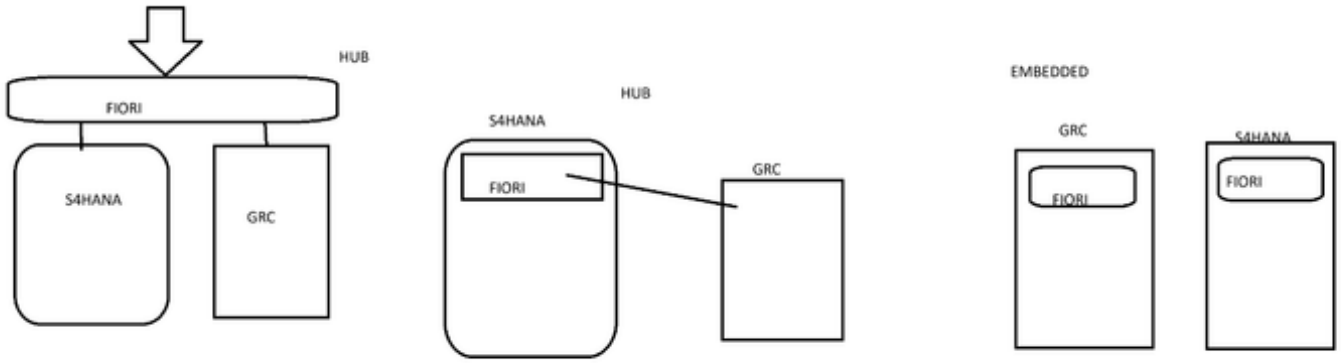
EX,, Access Request Workflow ------- > Process ID

- **New user Request**
- **Change user request**
- **Delete User Request**

GRC Fiori Integration:

/n/UI2/FLP – FIORI Launch pad

https://blogs.sap.com/2020/06/30/sap-fiori-solution-for-sap-access-control/



Imp Steps under User Provisioning:

- Role Management
- User Provisioning
  - BAdIs for User Provisioning
  - Generate BRF+ Function for Access Rules-to-Role Mapping
  - Maintain Settings for HR Trigger
  - Maintain Service Level Agreements
  - Define Request Type
  - Maintain Priority Configuration
  - Define Employee Types
  - Maintain Number Range Intervals for Provisioning Requests
  - Define Number Range for Provisioning Requests
  - Maintain End User Personalization
  - Configure Attributes for Role Search Criteria in Access Requests
  - Maintain Provisioning Settings
  - Maintain User Defaults
  - Maintain Review Rejection Reasons
  - Maintain Password Self Service
  - Maintain CUA Settings
  - Maintain Provisioning Settings for IDM
  - Activate End User Logon
  - Simplified Access Request
  - Fiori Access Request

Workflow Configuration:

Access Control ----- > Workflow for AC ------- > Maintain MSMP Workflows

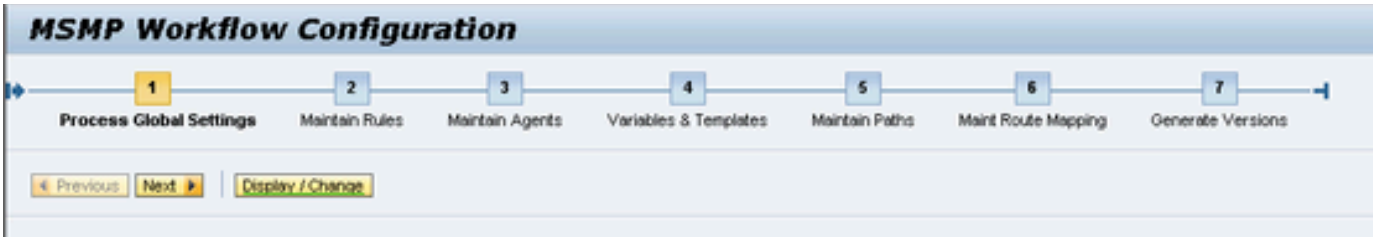
MSMP – Multi Stage Multi Path

Process IDs (Workflow types) – 11 Types of workflows

Select workflow process

Process ID	Process Description
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SAP_GRAC_ROLE_APPR	Role Approval Workflow
SAP_GRAC_SOD_RISK_REVIEW	SOD Risk Review Workflow
SAP_GRAC_USER_ACCESS_REVIEW	User Access Review Workflow

Workflow configuration is 7 step processes.

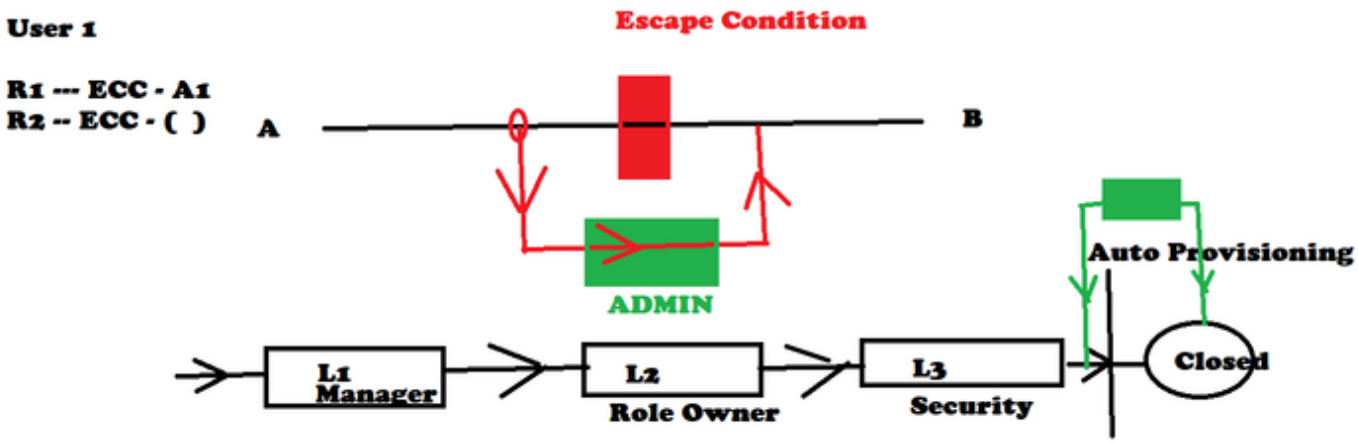


1. Process Global Settings:

- Decide which workflow type/process you are working on and that must be selected.
- Decide which mode to be used – **DISPLAY/CHANGE**
- Notification Settings - **Request Submission** and **End Of the request**
- Escape Conditions

Escape Conditions:

- **Approver Not Found** Ex, Role Owner not defined in BRM for a role/ manager is locked/terminated
- **Auto Provisioning Failure** – Ex, Broken RFC / Backend System Down

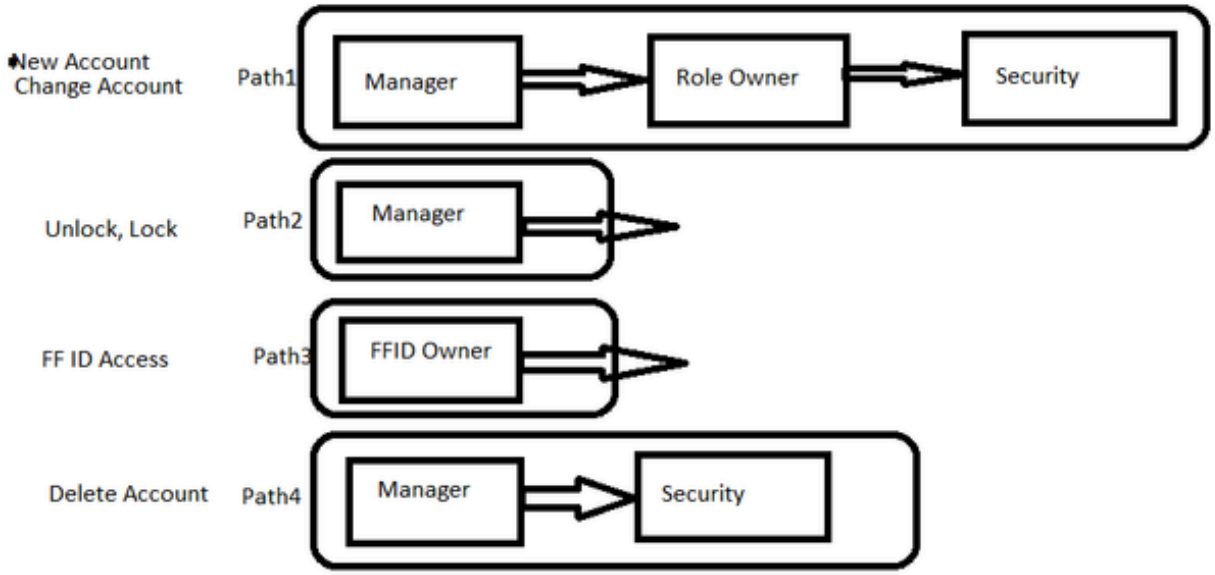


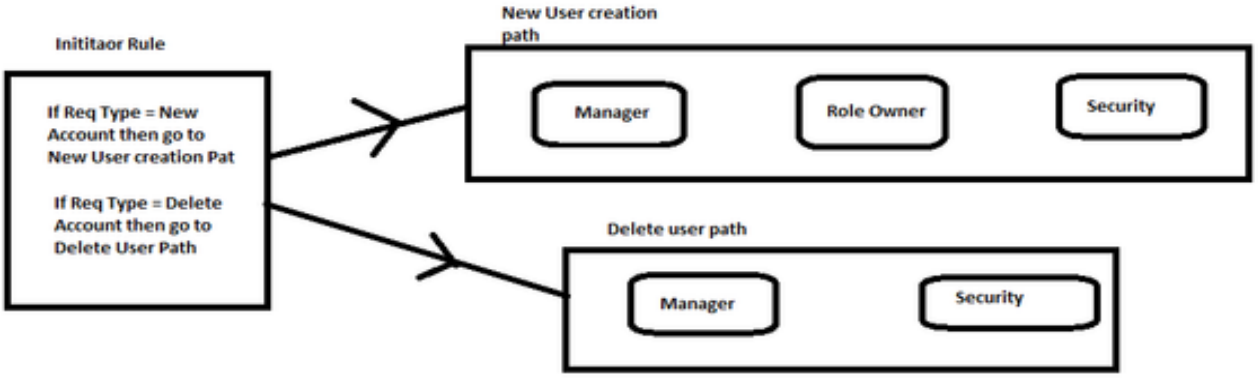
1. Maintain Rules

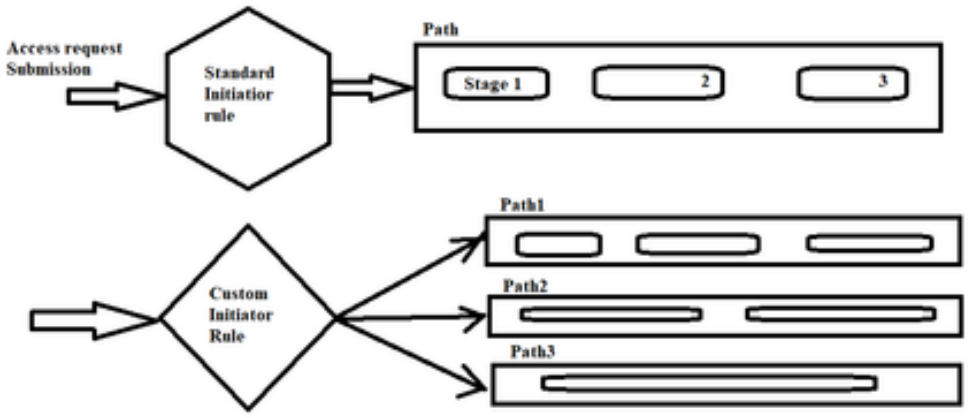
Rule Kinds: (Kinds of Rules)

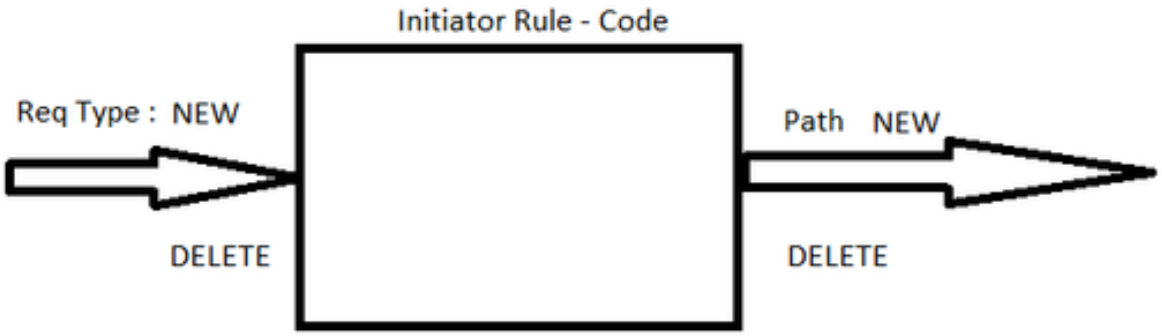
- Initiator Rule (1) ----- > Determines which **Path** to be followed.
- Agent Rule (many)------ > Determines **person (Approver/Requestor/User)** in the respective stage who is in charge.
- Routing Rule (many) ------ > Determine **Detour Path** (alternate path) to be followed when some condition is satisfied.
- Notification Rule (1) ------ > Determines **what Notification** to be sent, based on the action/event (Request submission, Approve, Reject, and Forward, on hold, End of the Request) taken by Approver (Manager/Role Owner/Security/FFID OWNER….)

Initiator Rule:

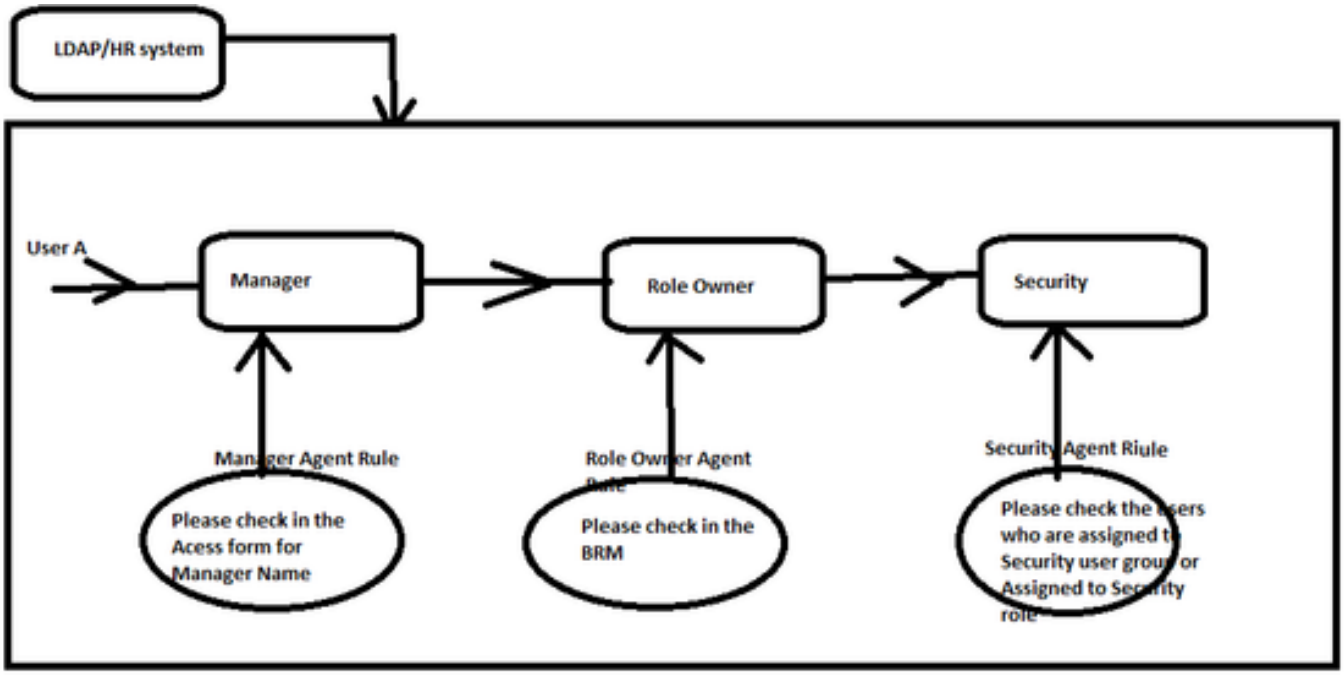

New Account
Change Account
Path1
Manager
Role Owner
Security
Unlock, Lock
Path2
Manager
FF ID Access
Path3
FFID Owner
Delete Account
Path4
Manager
Security


Inititaor Rule
If Req Type = New Account then go to New User creation Pat
If Req Type = Delete Account then go to Delete User Path
New User creation path
Manager
Role Owner
Security
Delete user path
Manager
Security


Access request Submission
Standard Initiatior rule
Path
Stage 1
2
3
Path1
Custom Initiator Rule
Path2
Path3

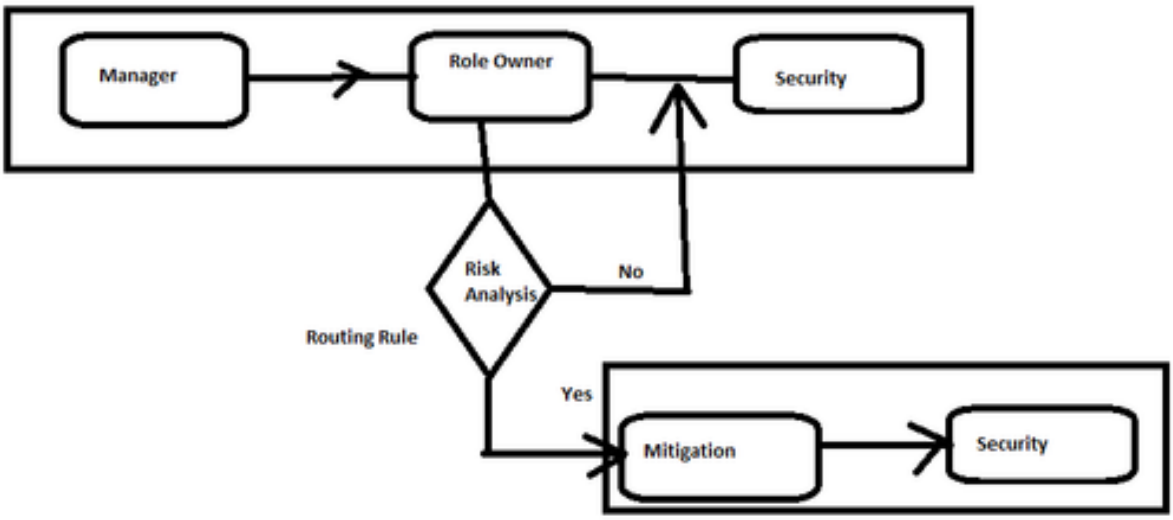


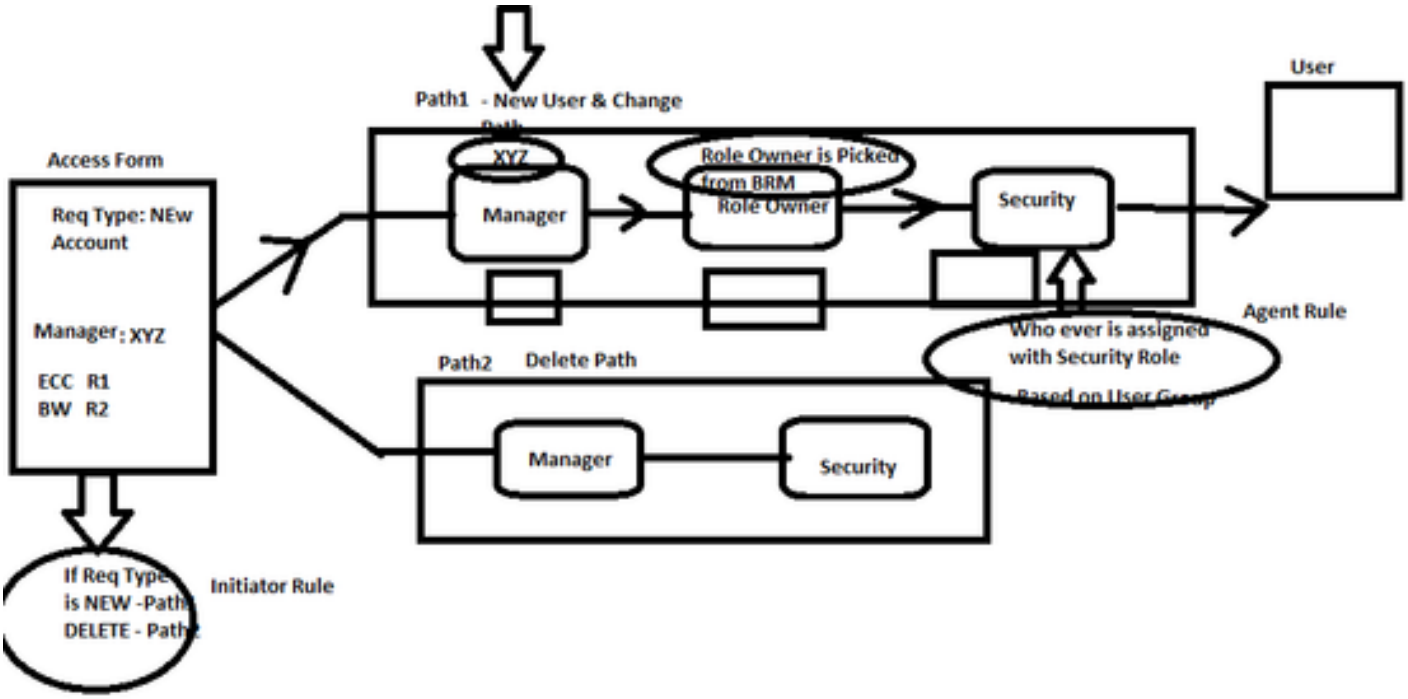
Agent Rule:



Routing Rule: (Detour/Alternate Path)
Conditions/Obstruction:

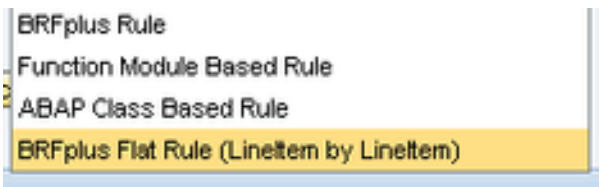
1. User has a Risk
2. Role Owner is missing.
3. Training Verification





Rule Types:

It indicates how the rule is created or can be customized.



Custom rules can be created in the below T-codes:
BRF+/BRFPLUS ---- > BRF Plus Rule & BRF Plus Flat Rule ----- > ABAP Developer/GRC Administrator
SE38/SE24 ----- > ABAP Class Based ----- > ABAP Developer
SE37 ------> Functional Module Based ----- > ABAP Developer

- All SAP Standard Rules are **Function Module Based** Type
- When Standard rules are not satisfying your requirement then we need to customize the rules, this step is mandatory.
- If you decide to use standard rules, this step can be skipped.
- Define the **Initiator rule** & **Notification Rule** for the workflow in this step

1. Maintain Agents

To customize Agent rules, when you don't want to use the existing Agent rules in Step 2

Example:
Teams/Departments – BASIS, Security, ABAP, FIN, HR, MM, Sales
Every team has Dedicated Manager

BASIS – M1
Security – M2
ABAP – M3
FIN – M4
When any user is submitting access form, 1st level should go to team manager.

Agent Types:



1. **Directly Mapped Users** – Directly informing to system who is approver by mentioning user id or name.
2. **PFCG Roles** – Whoever is assigned to a particular role.
3. **PFCG User Groups** – Who ever belongs to a particular user group
4. **GRC API rules** – Whoever is defined in **Step 2 of MSMP** as Agent Rules

- API Rule :Security --- > Whoever is defined as Security lead under Access COntrol Owners
- User Group : ZSECURITY1 ----> Whoever is assigned with User Group SECURITY
- Role Based: ZSECURITY2 ---> Whoever is assigned with SECURITY Role
- Directly Mapped User: ZBASIS--- A1, A5, A7

Agent Purpose:
- Approval
- Notification



1. **Variables & Template**

Customize templates… Mail formats…
Already we have predefined mail formats for every action/event.
SE61 – Customize notification templates

Workflow ------ > Initiator ------ > Rule Result ------ > Path ----- > Stages 1, 2, 3

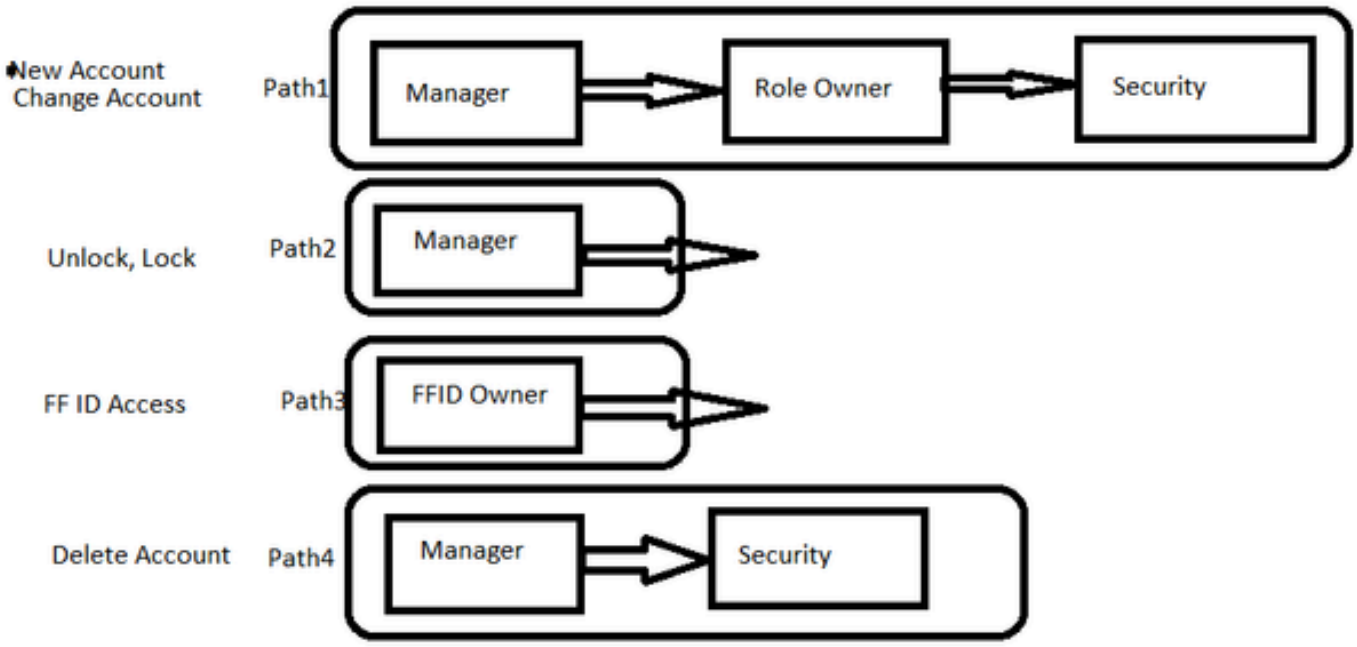
Ex,, **Access request Approved Template**

Access request "number" submitted by "Requestor" is approved in "Stage Name" and now is pending in "Stage Name" Stage "XYZ"

1. **Maintain Paths**

Only one path is available which is called **GRAC_DEFAULT_PATH**
This path has 3 stages.

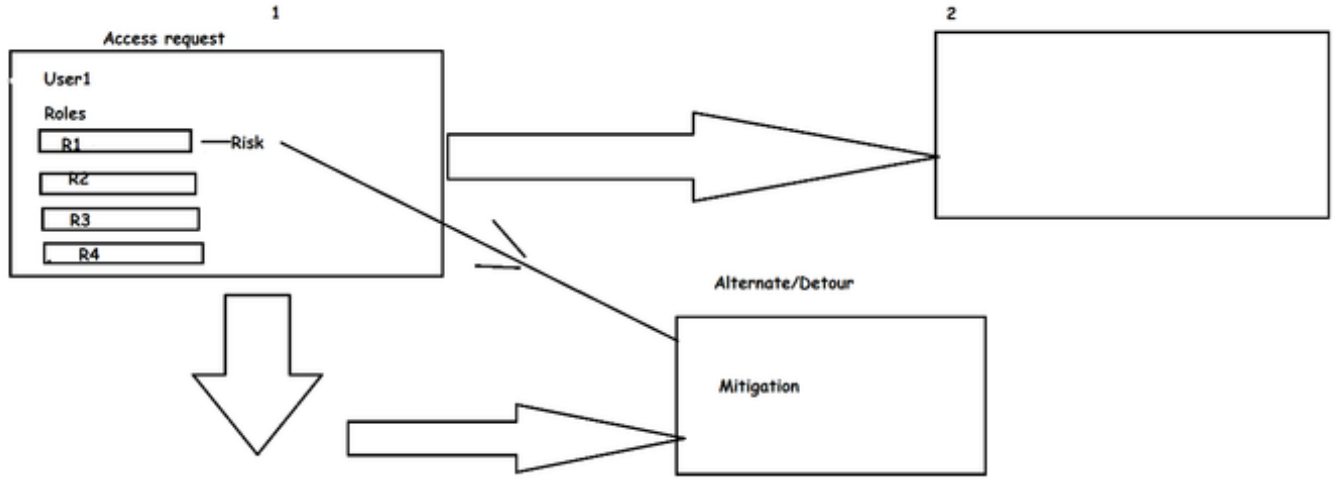
1. Manager
2. Role Owner
3. Security



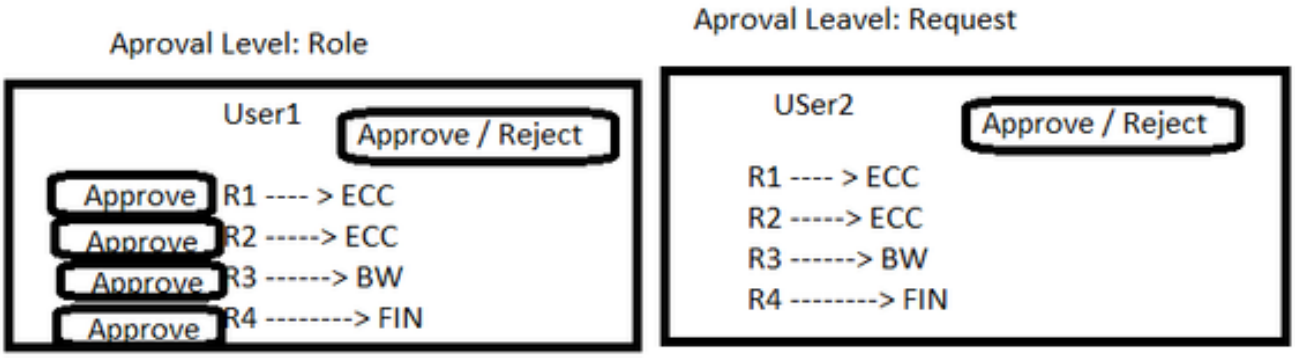
We have many request types like

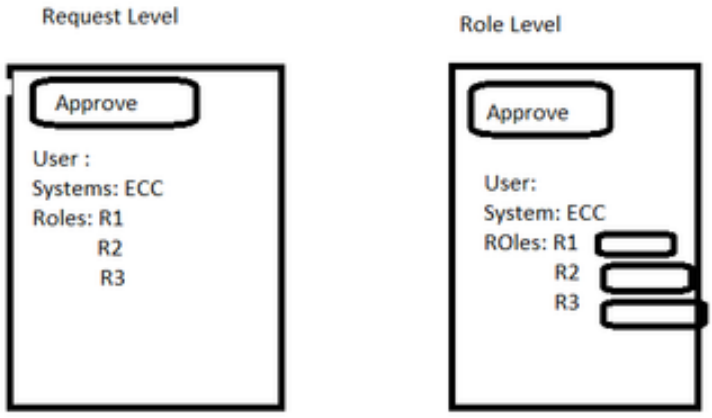
- New User Request
- Change user Request
- Delete User Request
- Lock Account
- Unlock Account
- FF ID Access

We need multiple paths as we have multiple request types



Modify Task Settings:





1. **Maintain Route Mapping:**

- Mapping Paths with Initiator rule
- Mapping Alternate path (detour path) with Routing rule

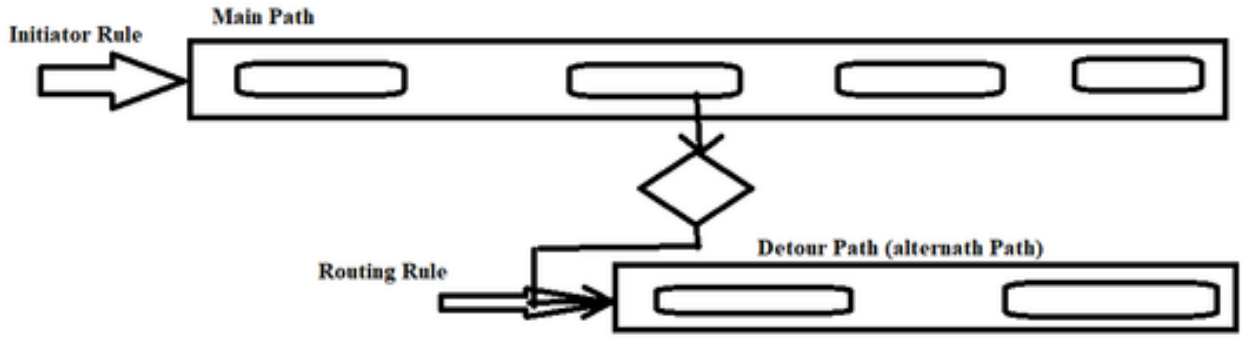
No of Paths: 1
Initiator Rule: Standard

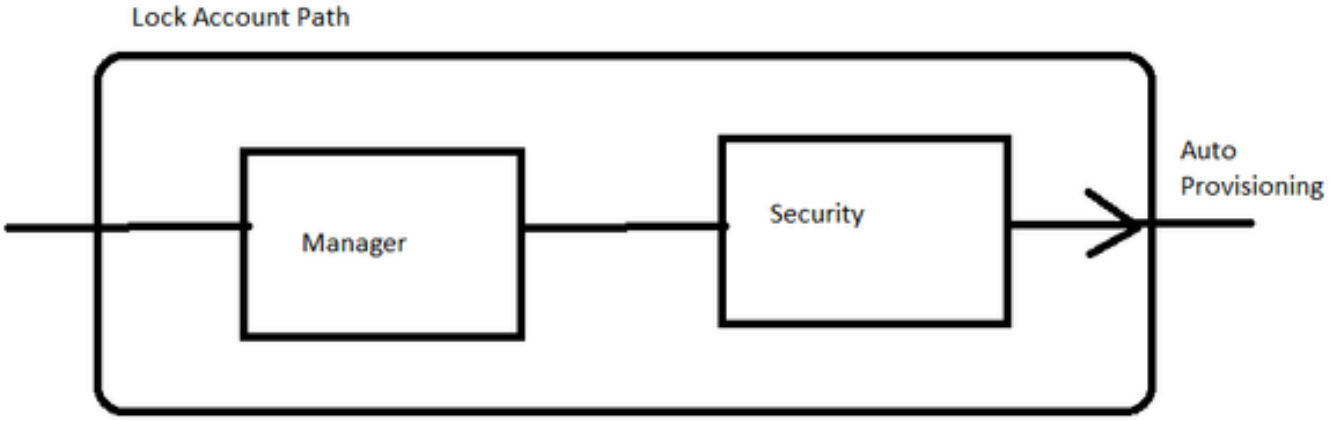
No of Paths: > 1
Initiator Rule: Custom

Default Initiator rule

Rule Result 1

Custom Initiator Rule		
Rule Result 1		
Rule Result 2		
Rule Result 3		
Rule Result 4		
Rule Result 5		





BRF+: Business Rules Framework

It can be accessed by the T-code.

- BRF+
- BRFPLUS

Or

SPRO --- > GRC ----> AC -----> Workflow for AC ------> Define BRF

Purpose: To customize Rules (**Initiator**/Agent/Routing/Notification)

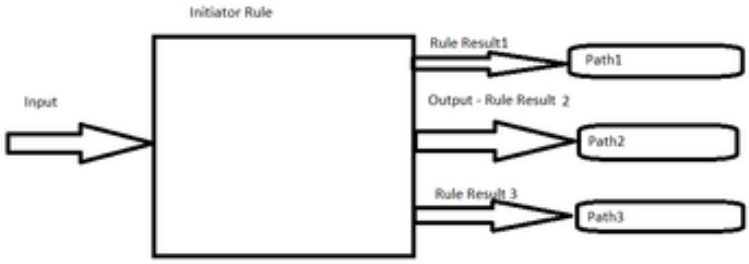
- We are customizing the **Initiator rule** since standard default Initiator rule has only one rule result which can be linked to one path only.
- We need multiple rules results under the Initiator rule so that we can link them up with multiple paths.
- Hence, we are going to BRF+ application to create Custom Initiator Rule with Multiple Rule Results

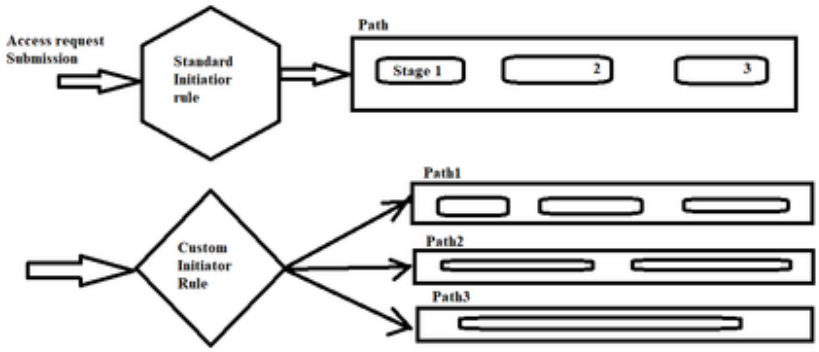
Default Initiator rule

Rule Result 1

DEFAULT PATH

Custom Initiator Rule		
Rule Result 1	Path 1	
Rule Result 2	Path 2	
Rule Result 3	Path 3	
Rule Result N	Path N	





How to create a custom Rule?

1) BRF+ Application ----- > BRF+ or BRFPLUS
2) ABAP Class based ---- > SE38/SE24
3) Functional Module Based ----- > SE37

Request Type	Custom Initiator Rule		
New Account	MRF_NEW	Path1	NEW USER REQUEST
Change Account	MRF_CHANGE	Path2	CHANGE USER REQUEST
Delete Account	Rule Result 3	Path3	DELETE USER REQUEST
Lock Account	Rule Result 4	Path4	Lock REQUEST
Unlock Account	Rule Result 5	Path5	Unlock USER

Initiate the use of BRF+ from SPRO and create BRF+ Application:

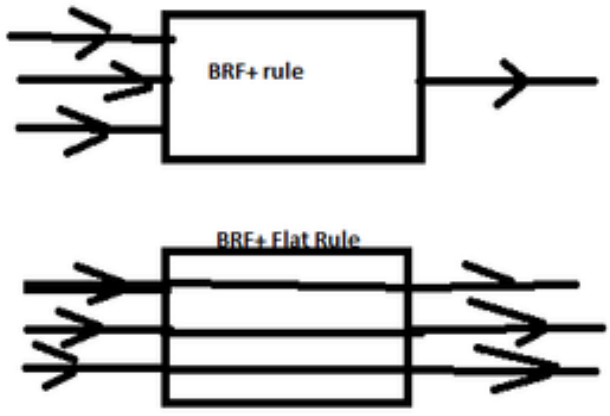
Launch the BRF+ from SPRO:

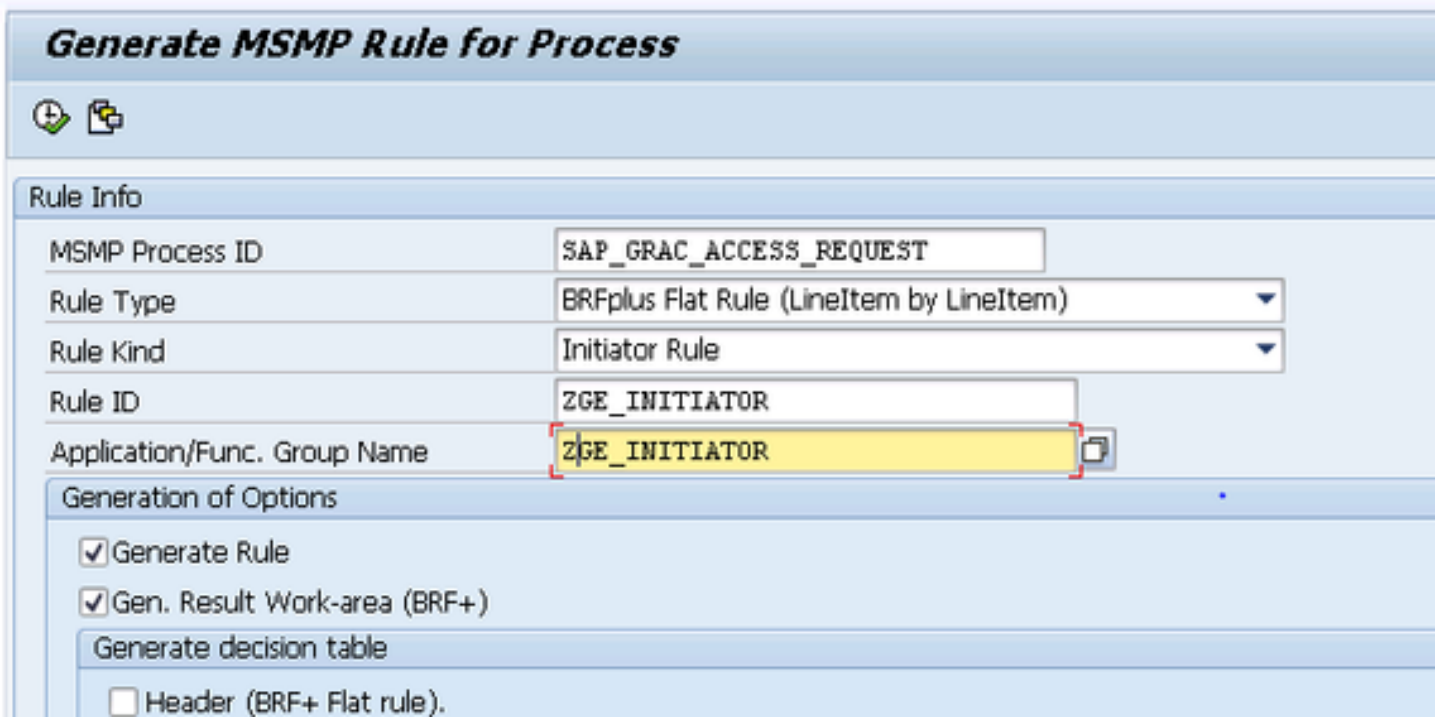
GRC ---- > AC ------ > Workflow for Access Control ------ > **Define Business Rule Framework**

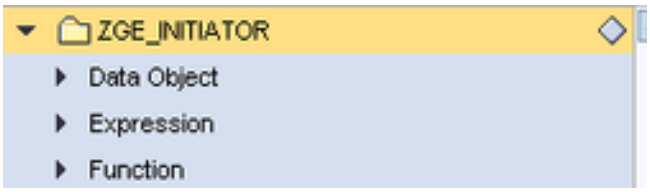
1. Need to create Application under BRF+
2. Application can be created from the below path in SPRO

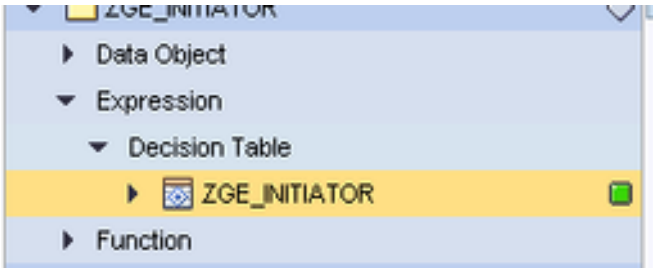
Path: GRC ---- > AC ------ > Workflow for Access Control ------ > **Define Workflow-Related MSMP Rules**

What is the difference between BRF+ rule and BRF+ Flat Rule (line item by line item)?

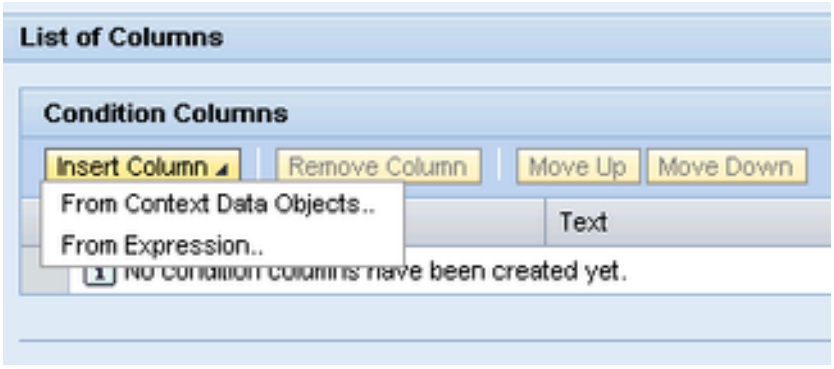


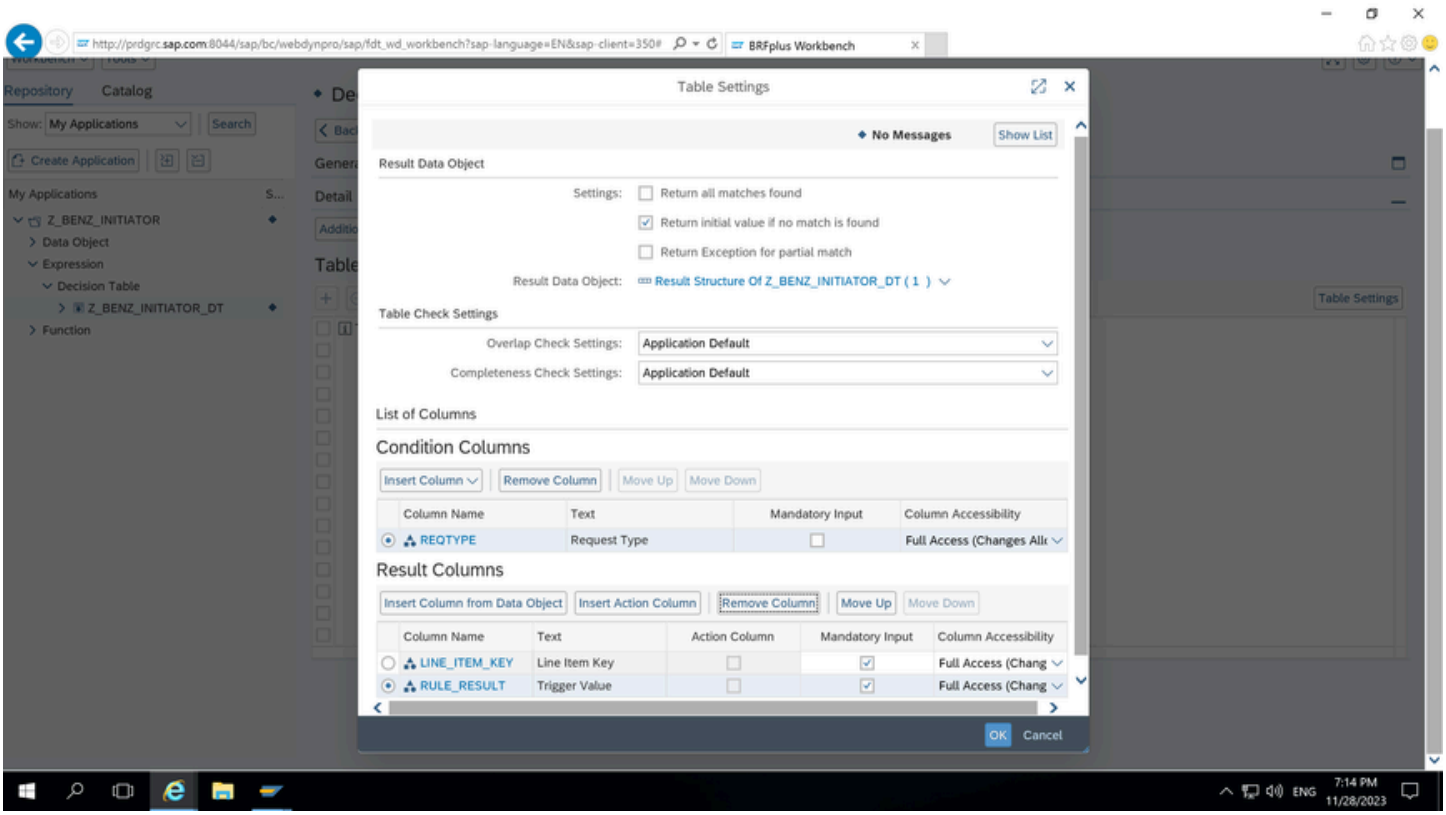


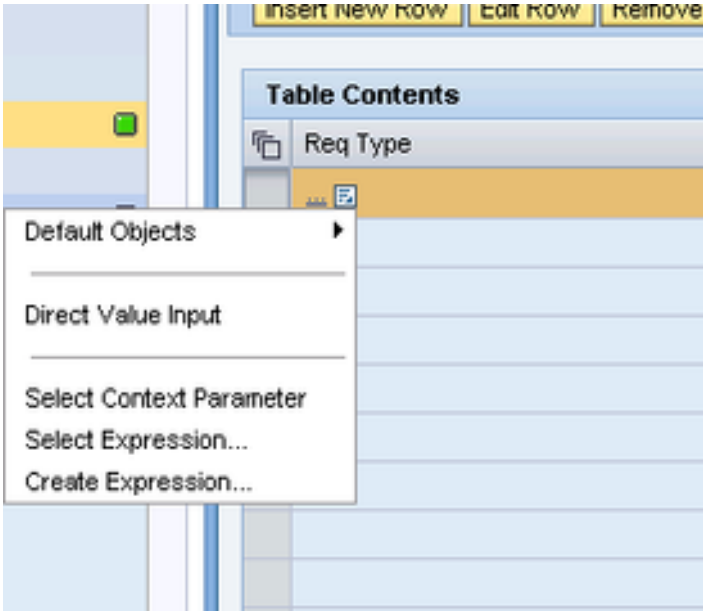


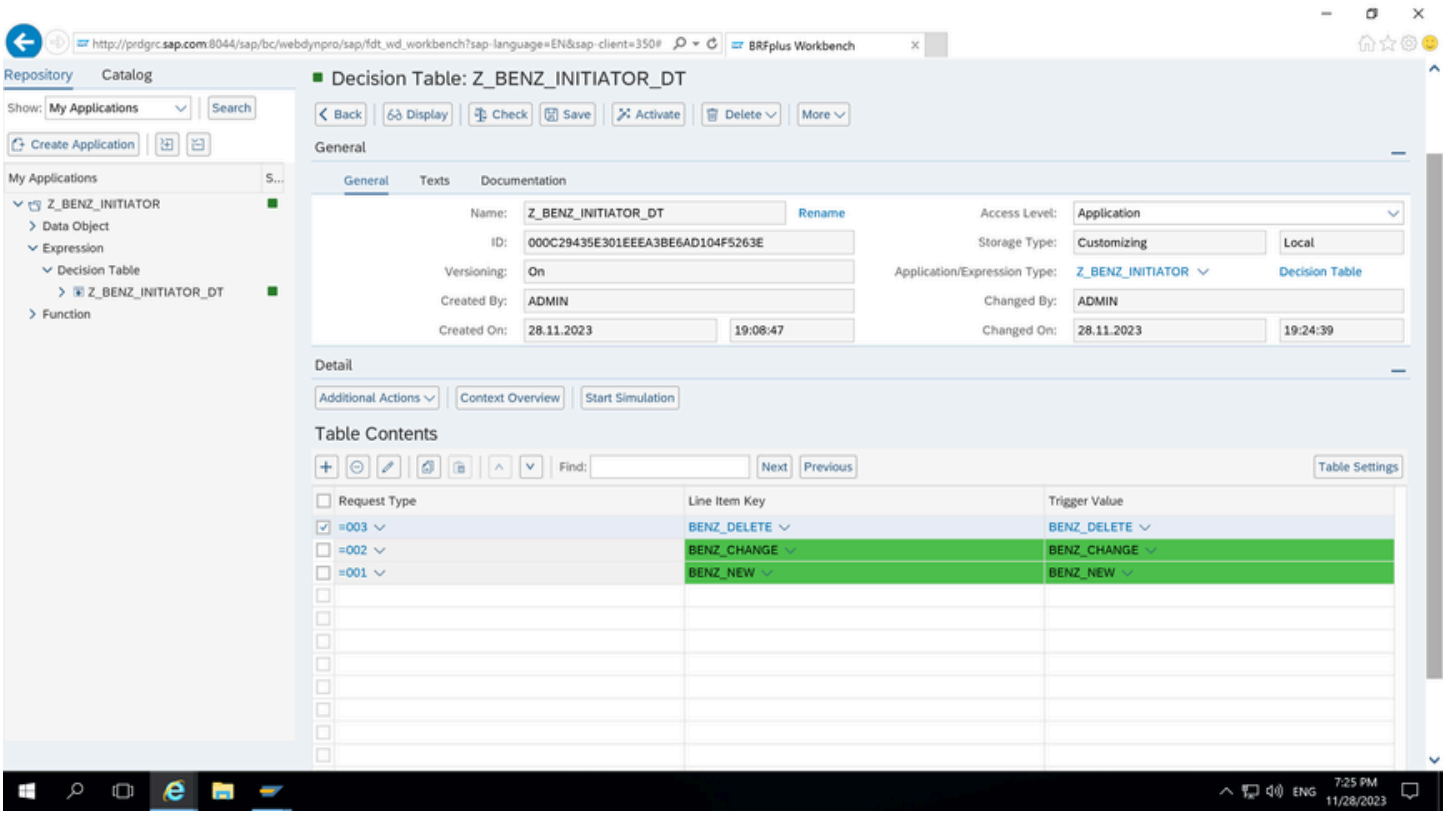


Choose CONTEXT DATA Objects below



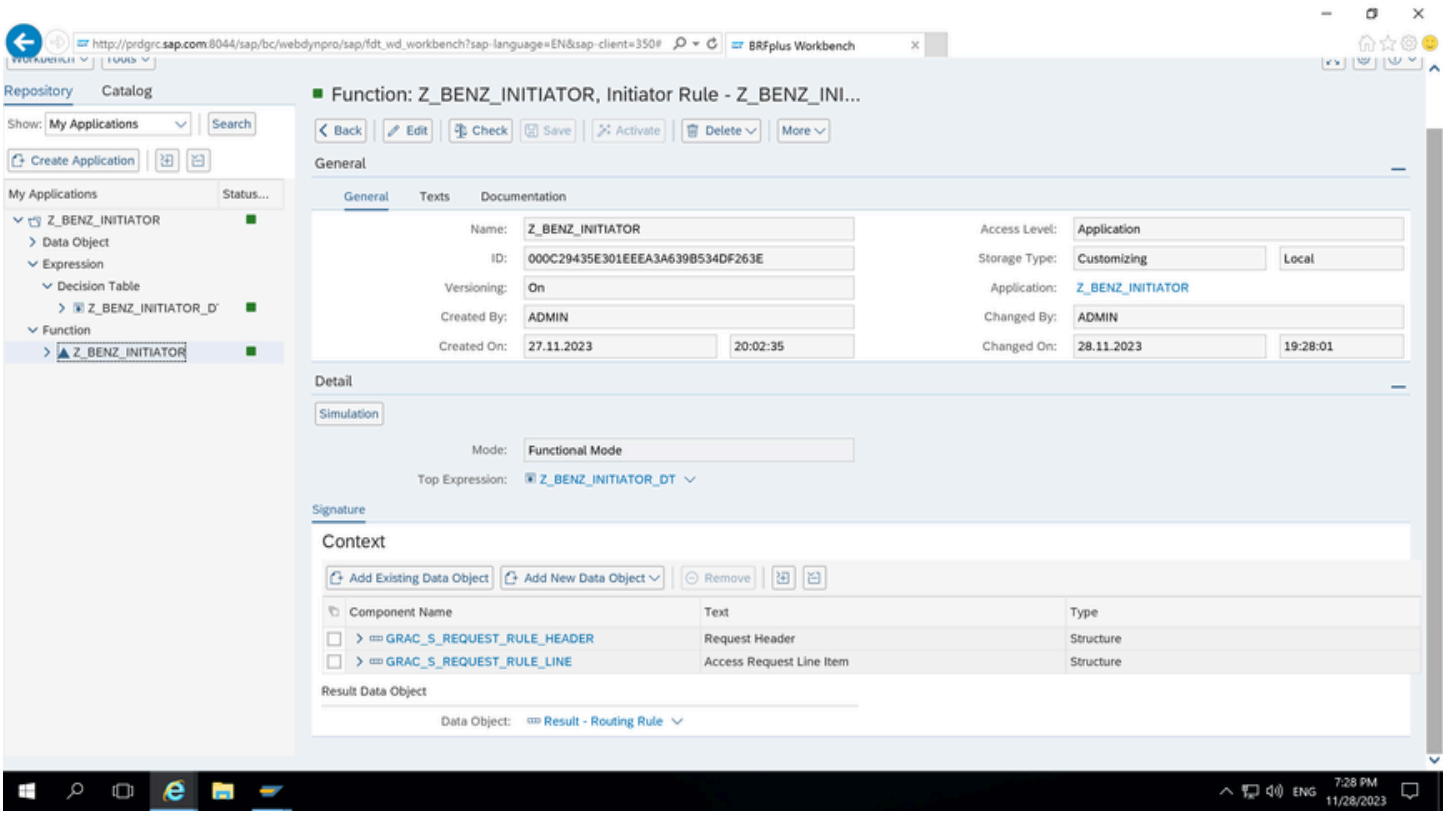

BRFplus Workbench
Repository
Catalog
Show: My Applications
Search
Create Application
My Applications
Z_BENZ_INITIATOR
Data Object
Expression
Decision Table
Z_BENZ_INITIATOR_DT
Function
Table Settings
No Messages
Show List
Result Data Object
Settings:
Return all matches found
Return initial value if no match is found
Return Exception for partial match
Result Data Object:
Result Structure Of Z_BENZ_INITIATOR_DT (1)
Table Check Settings
Overlap Check Settings:
Application Default
Completeness Check Settings:
Application Default
List of Columns
Condition Columns
Insert Column
Remove Column
Move Up
Move Down
Column Name
Text
Mandatory Input
Column Accessibility
REQTYPE
Request Type
Full Access (Changes Allx
Result Columns
Insert Column from Data Object
Insert Action Column
Remove Column
Move Up
Move Down
Column Name
Text
Action Column
Mandatory Input
Column Accessibility
LINE_ITEM_KEY
Line Item Key
Full Access (Chang
RULE_RESULT
Trigger Value
Full Access (Chang
OK
Cancel
ENG
7:14 PM
11/28/2023


Insert New Row
Edit Row
Remove
Table Contents
Req Type
Default Objects
Direct Value Input
Select Context Parameter
Select Expression...
Create Expression...



Direct Value Input

Activate FUNCTION.
Insert Top expression – Decision Table

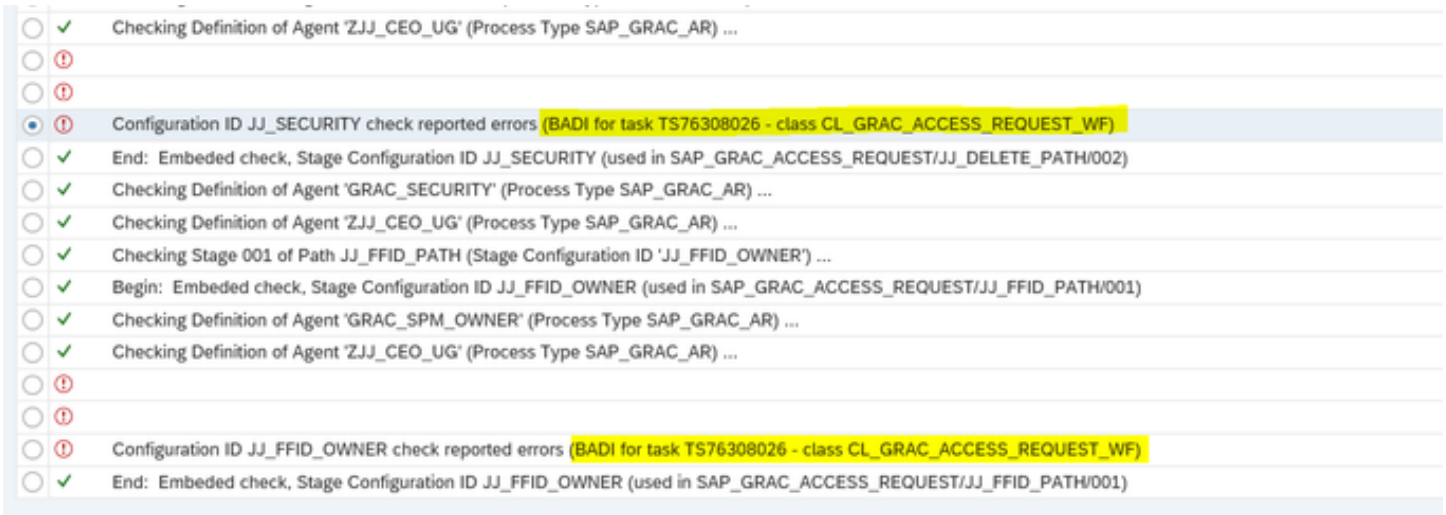


BRF+
A = B (decision Table)

MSMP
B = C

A = Request Type
B = Trigger Value
C = Path

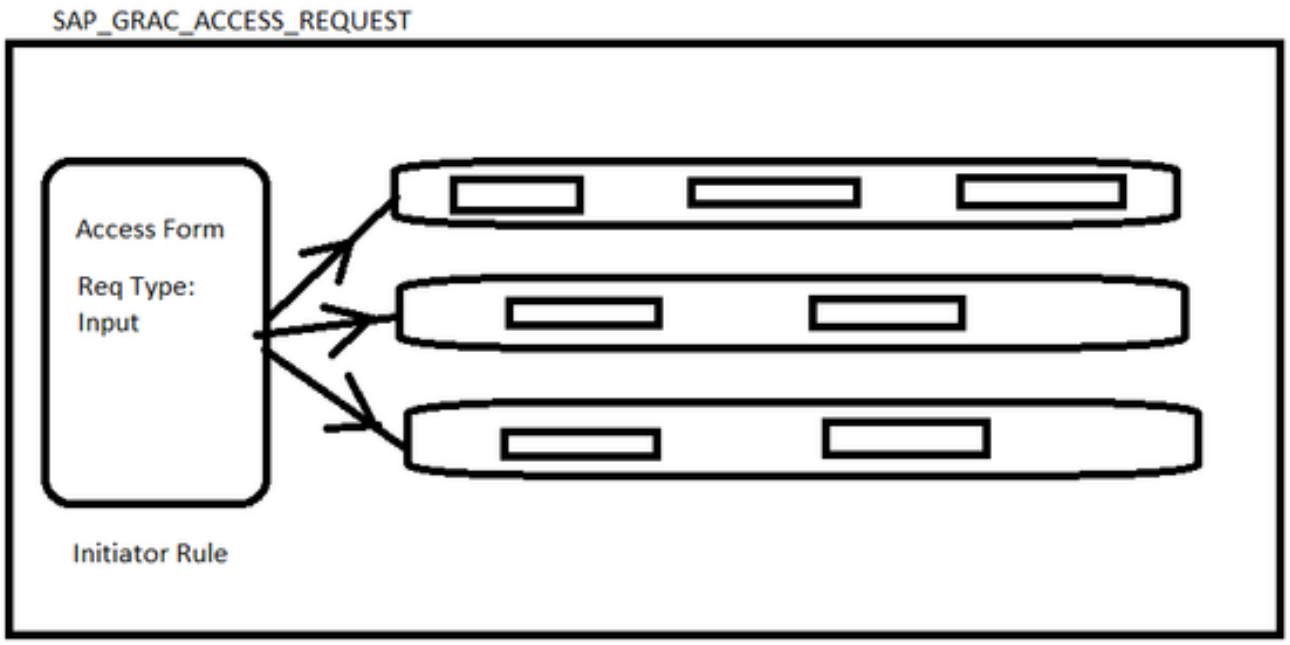
<u>Save and Simulate:</u>



<u>Access request Workflow</u>

ZTYT_INITIATOR:
REQ TYPE ------- > Rule Result ------- > Path ---- > Stages
001------- >JJ_NEW------- >JJ_NEW_USER_PATH
003 -------- >JJ_DELETE ----- > JJ_DELETE_PATH
004------- >JJ_LOCK------- >JJ_LOCAK_PATH
005------- >JJ_UNLOCK------- > JJ_UNLOCK_PATH
006------- >JJ_FFID------- >JJ_FFID_PATH

Why is it called **MSMP**? (Multi Stage Multi Path)
Earlier in GRC 5.3, every request type has dedicated Initiator
GRC 5.X ---- > Multistage Single Path
GRC 10.X, 12.X ----- > Multistage Multi Path
In GRC 10, 10.1, 12, entire workflow (no of paths) has one initiator.

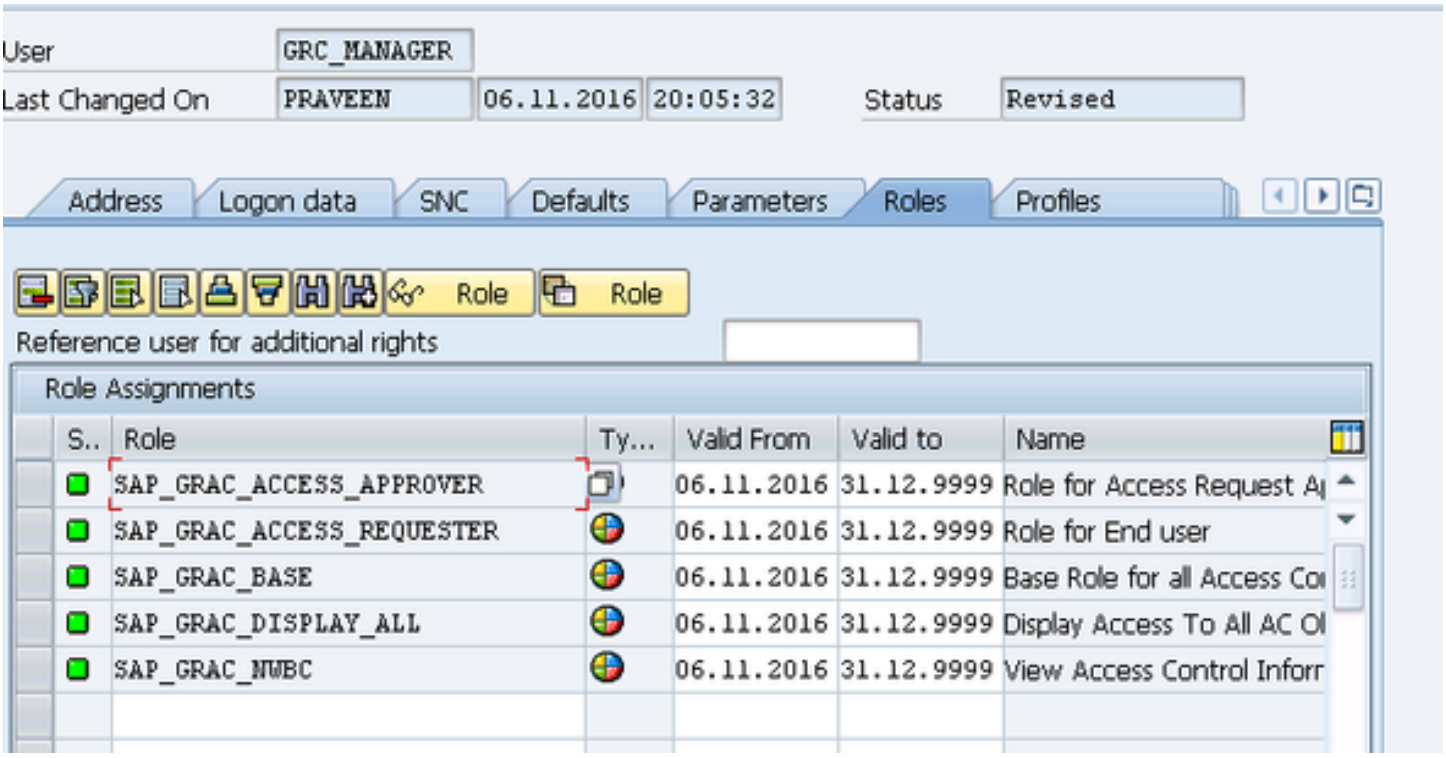


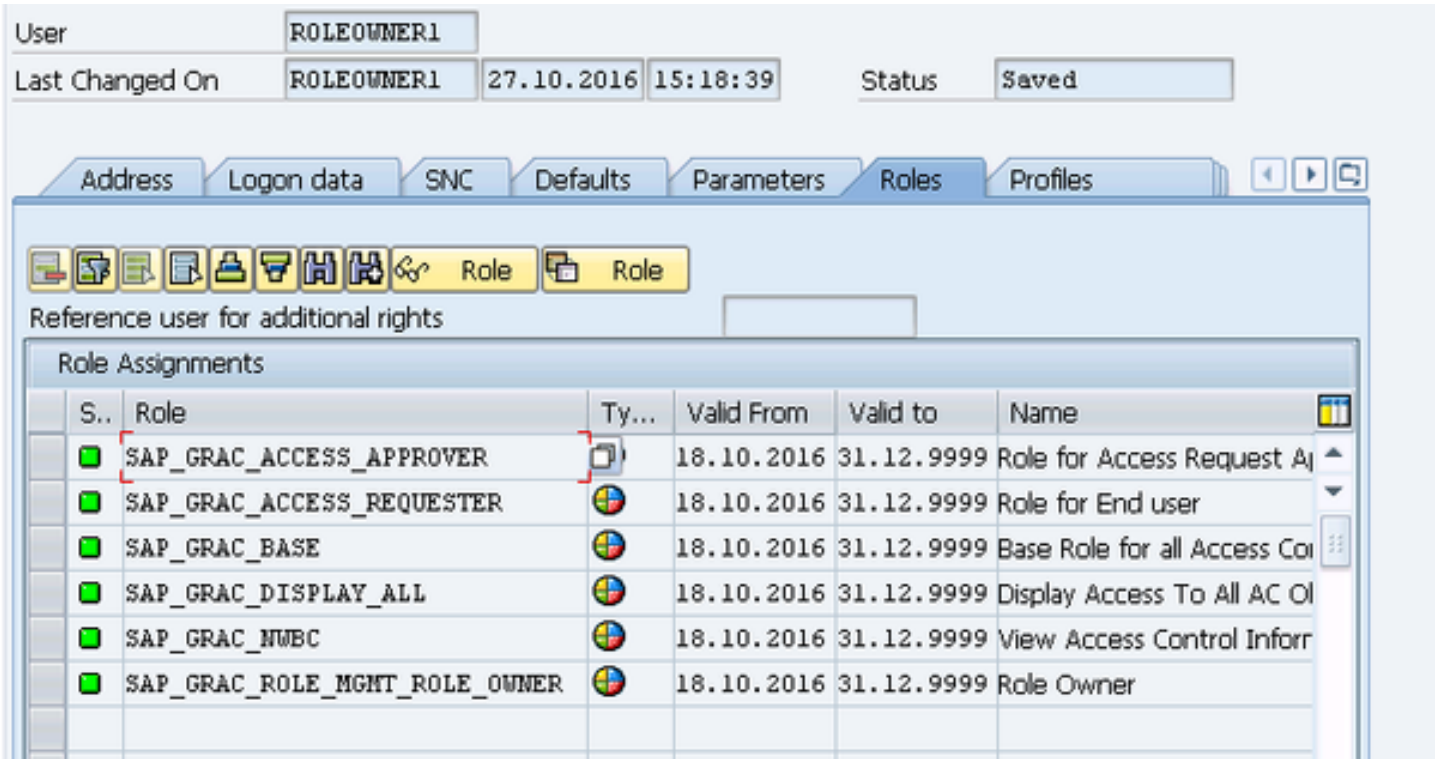
Testing:
Testing can be divided into 3 areas.

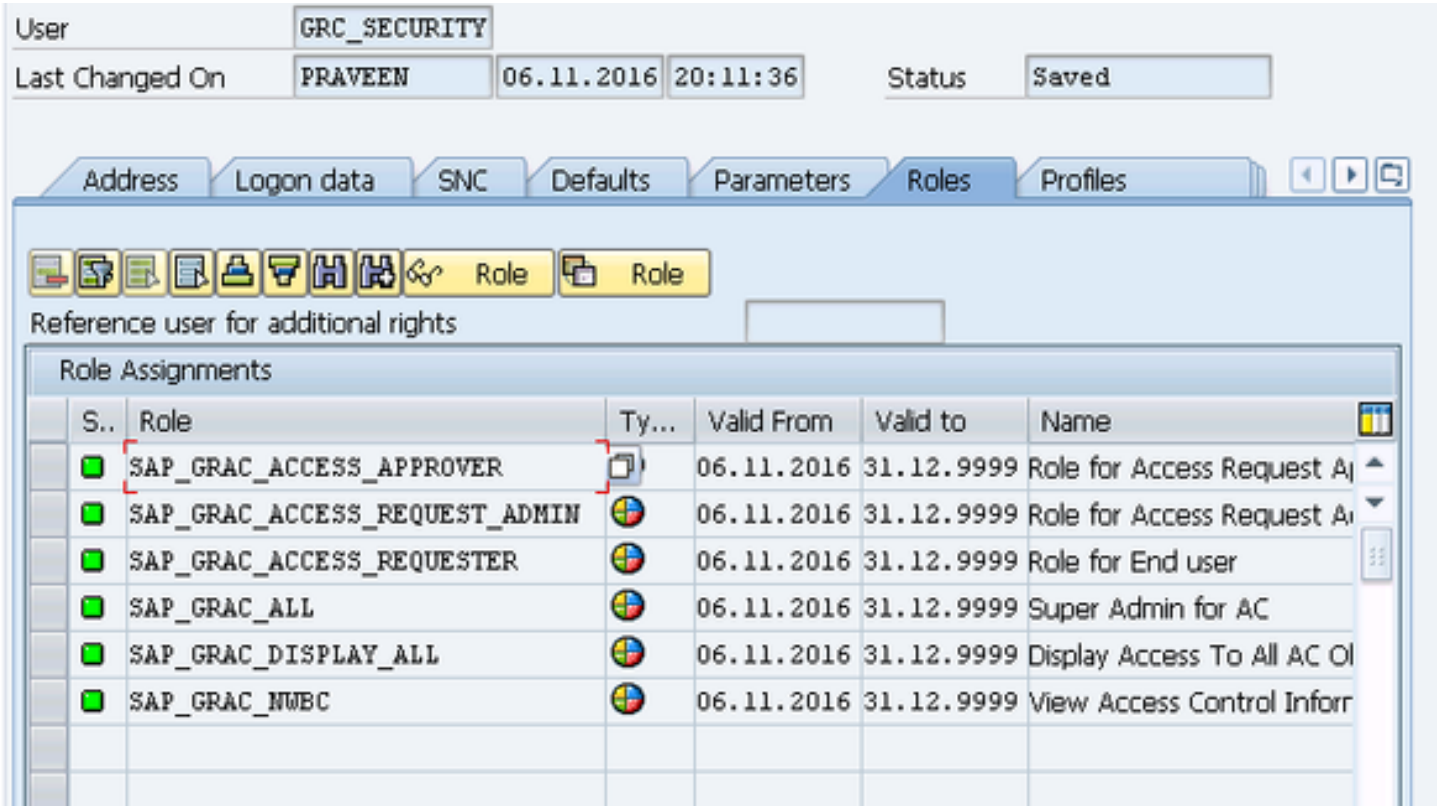
- Request Submission – New Account, Change, Delete, Lock, unlock.
- Approval Stages (MSMP workflow)
- Auto Provisioning

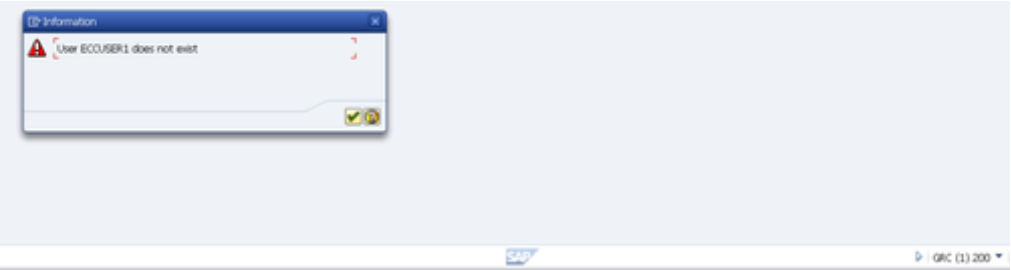
Users – Manager, Role Owner and Security, FFID OWNER, CEO Risk Owner – have to exist in GRC system with relevant access to approve the requests in their respective stages.
MANAGER – Stage1
ROLEOWNER – Stage 2
SECURITY – Stage 3

Create PG_USER2
Delete PG_USER1
Ensure roles have approvers defined and check the settings for the role so that it should be visible for the user when filling in the access form.









SLG1: Troubleshooting T-code
GRFNMW_DBGMONITOR_WD: Troubleshoot MSMP Workflow
LPD_CUST: Customizing the NWBC Page – launch Pad
GRFNMW_DEBUG: Debug MSMP Workflow

Table:

GRFNMWCNVERA – To check latest active version number and who generated it
GRFNMW* Tables

UAR: **User Access Review** is used by managers and role approvers to automate the periodic process of reviewing and reaffirming end user role assignments. The system notifies managers to review their direct reports' access to applications. ... Approvers either approve or remove the roles that are assigned to each user.

Employee – Joined in 2020 -- > Job 1
2023 ---- > Job2
Today – 2026 ---- > Job 3
User ID: ----- > Manager or Role Owner should do review – 6 Months
Job1 --- > Role 1 --- Officer
Job 2 --- > Role 2 --- Supervisor
Job 3 ---- > Role 3 --- Manager

Segregation of Duties Review (SOD Review)

Segregation of Duties Review is a process where the system checks periodically for any risk and violations associated with a user or functions. This functionality can be used during the **initial clean-up of risk violations** as well as a long-term strategy to review and affirm **previous Mitigation assignments**.

When SOD review is performed, it generates requests automatically, based on the organization's internal policy. SOD review provides Workflow Based review and approval process.

Following paths in my MSMP Workflow

If User is **Employee** and **New user** – Path1 (2 Stage Approval)

If user is **Contractor** and **New user** – Path2 (3 Stage Approval)

Upgrade:

10/ 10.1 ----- > 12.0

10 & 10.1 Support is officially stopped from Jan 2021

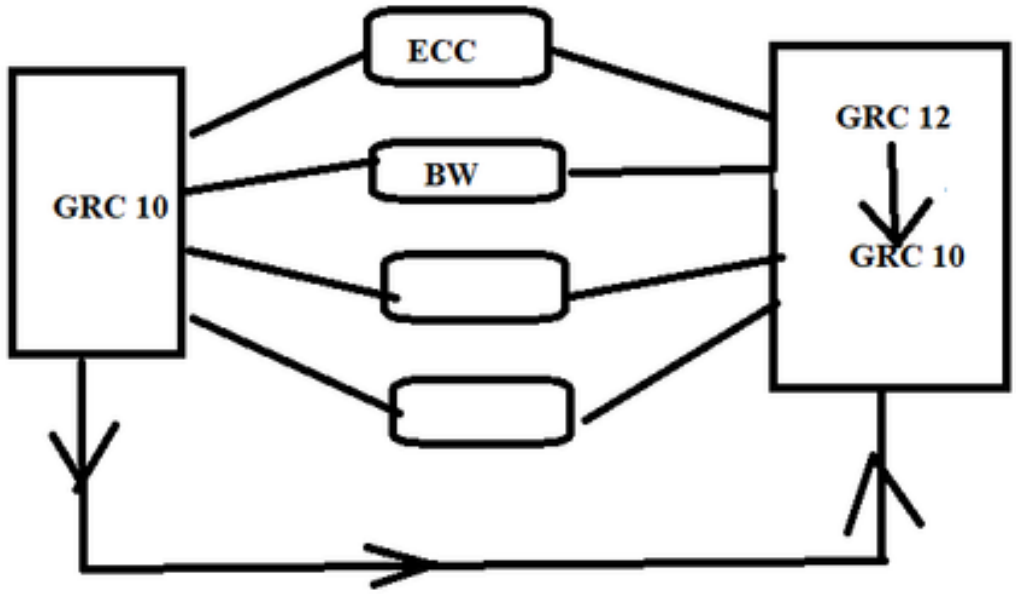
Support pack Upgrade ---- > GRC 12 --- > SP10 to SP31

GRC 12 ----- > GRC 2026 (version Upgrade)

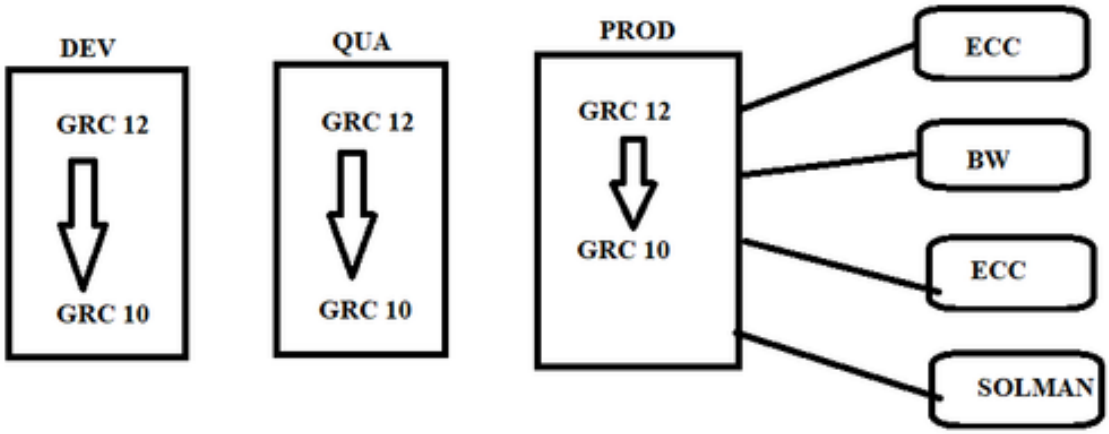
The latest Support Package (SP) level for **SAP GRC Access Control 12.0** is **SP31**, running on SAP NetWeaver 7.52.

New Version ---- > SAP GRC 2026

Scenario 1: GRC 10 is on old SID (Old server) and GRC 12 will be on new server with new SID.



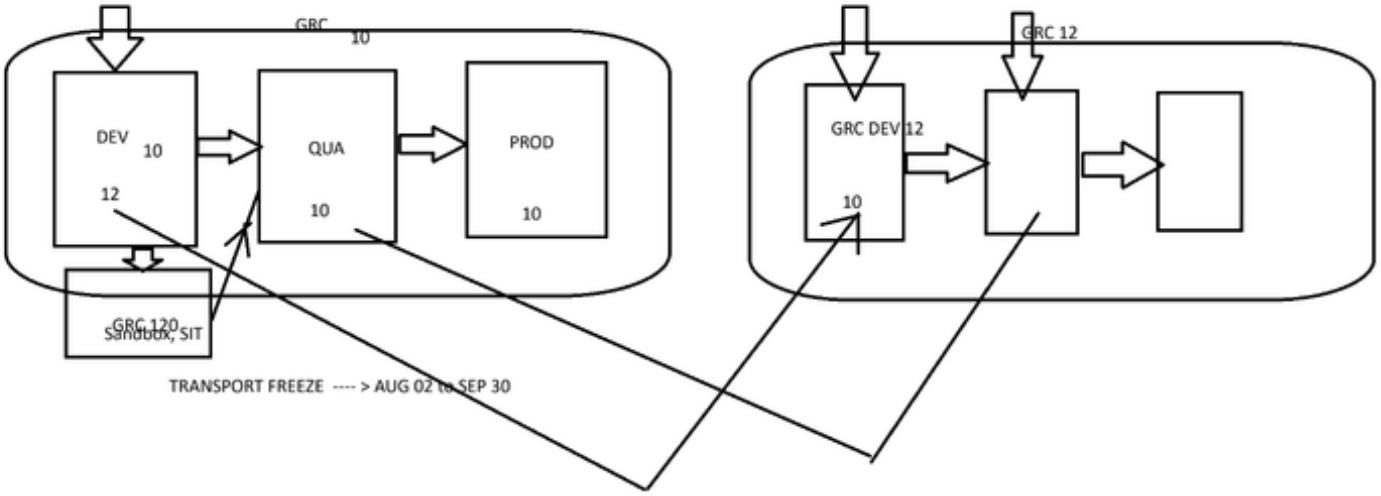
Scenario 2:



Jan 2026 -- > Dev Is upgraded from 10 to 12/ GRC 12 SP Upgrade 10 to 31

Mar 2026 --- > QAU

May 2026 --- > PROD --- Go Live



End user --- > Request submission, Req Status

Manager --- > Approve, Reject Access Request

Role Owner --- >

Risk Owner

Mitigation Approver and Monitor

FF ID owner and Controller

BRF Plus Assignment:

Client does not have LDAP/AD/HR where manager Info is usually stored

Team	Manager
BASIS	M1
Security	M2
ABAP	M3
FIN	M4
HR	M5
Sales	M6

New /Change Access Request:

Manager --- > It must be manually filled in the form – Standard manager agent cannot be used. Hence need to create custom Manager Agent Rule which will fetch manager info from the above table.

Role owner

Security